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Matteo Luciardello Lecardi	10572870	matteo.luciardello@mail.polimi.it
Francesco Pezzi	10567735	francesco1.pezzi@mail.polimi.it
Josè Jorge Rodea Valdes	10769772	josejorge.rodea@mail.polimi.it
Lorenzo Voltini	10570379	lorenzo.voltini@mail.polimi.it
Luca Stefano Zaccanti	10583416	lucastefano.zaccanti@mail.polimi.it

ABSTRACT

The present report presents in detail the phase A feasibility study for a 24 Unit CubeSat for a scientific mission in the Jupiter planetary system. The CubeSat will carry two scientific payloads.

The objectives are to study the atmosphere of Jupiter's moon Europa, mapping the entire surface of the satellite to spot possible water plumes, analyzing their composition, and locate possible landing spots for possible future missions.

The mission is achieved with a low mass and 24 U size architecture, in the next chapters all the subsystems necessary for the mission will be defined and sized to ensure the feasibility of the mission.

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1. INTRODUCTION

The team identified its interest in Europa because ground-based telescopes on Earth, along with Voyager 1 and 2, Galileo spacecrafts and space telescopes, have increased scientists' confidence for an ocean under Europa's ice shell.

Since recent observations from Hubble Space telescopes and reanalysis of some data that Galileo spacecraft collected, specialists believe that it is possible that thin plumes of water are being ejected above Europa's surface.

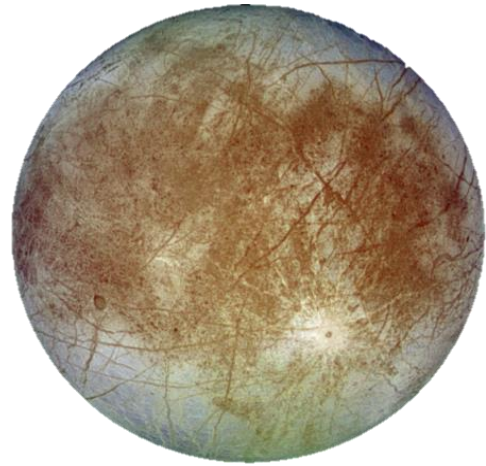


Figure 1.1 - Europa image from Galileo mission [1]

Considering that water is the major resource for life itself, that it could be useful for future human space missions on Europa, and taking into account that Europa's vast and unfathomably deep ocean is widely considered the most promising place to look for life beyond Earth, by investigating Europa we expect to gain the public and the youth attention, in order to increase Space Exploration interest.

2. MISSION GOALS

The mission goal is to send a satellite to Europa, one of Jupiter's Moons, to accomplish a scientific research.

The first goal is to map the surface of Europa: the objective is to elaborate a 3D physical-chemical model of Europa's icy soil, mapping 100% of surface at 1.2m per pixel. This goal will be achieved with the Chameleon Imager: a narrow angle camera manufactured by SCS space.

By doing that, the expected results are:

- to capture features of crust structure: valleys, mountain ranges, ridges;
- to access caves, ice sheets [12];
- to detect traces of sulfates, carbonates and organics from ocean [8];
- to detect potential landing spots for future research.

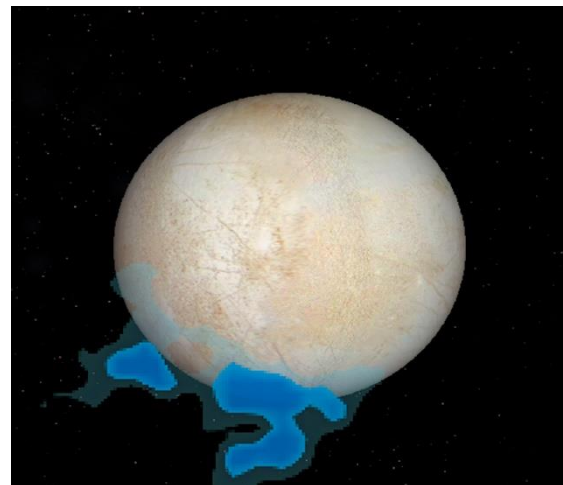


Figure 2.1 - Water plumes erupting from Europa's surface [2].

Then, the mission will conduct plumes and atmosphere investigation: the objective is to examine the chemical composition and the key organic compounds in Europa's plumes and atmosphere. This will be done with the MASPEX, a mass spectrometer specifically designed for looking for life-related compounds.

The expected results are:

- to intercept multiple plumes;
- to know the main chemical composition of the plumes;
- to know the chemical compounds present in Europa's atmosphere.

All the acquired data will be shared with Space agencies and Space companies, so that they will be able to conduct feasibility studies for future human and robotic exploration missions on Europa.

3. NOVELTIES ADDED

Rhadamanthus concept aims to effectively and affordably manage two Europa Clipper's scientific missions. The proposal is to detach the CubeSat so that it can operate surface mapping and plumes investigation with a camera and a mass spectrometer, flying closer to Europa and for longer than the original EC's mission designed path. The expected result is a higher resolution imaging (order of a few meters per pixel) along with a total surface scan, rather than the EIS system (EC correspondent payload which is expected to provide up to 90% of the surface scan at 50m spatial resolution). On the other hand, EIS's Three-dimensional views will allow surface heights measurements. Concerning mass spectrometry, the new mission design will allow more available time for analysis of the water vapours spreading in space (the same MASPEX payload will be adopted).

After a discussion with the actual NASA teams working on EIS and MASPEX, it will be established if the original EC design is still recommended, offering reliability advantages, or if Rhadamanthus' sampling will be capable of replacing (at least partially) these Europa Clipper's payloads. The downlink performance is the main criticality to focus on.

4. FROM MISSION GOALS, FUNCTIONALITIES AND REQUIREMENTS TO INSTRUMENTS

The primary objective of Rhadamanthus mission is to send a 24 U satellite to Europa, with 2 scientific payloads, to accomplish scientific research.

4.1 MISSION OBJECTIVES

1. Surface mapping: build a precise topographic map of Europa's surface.
2. Plumes' investigation: analyze potential water plumes that were observed coming out from Europa.

	GOALS	OBJECTIVES	INSTRUMENTS
Surface mapping	Elaborate a 3D physical-chemical model of Europa's icy soil, mapping 100% of surface at 1.2m per pixel.	Hyperspectral linescan imaging	Chameleon Imager
Plumes' investigation	Analyze potential water plumes coming out from Europa and Europa's atmosphere	Analyze the chemical compounds of the atmosphere and their quantities	MAss Spectrometer for Planetary EXploration (MASPEX)

Table. 4.1 – Mission goals and objectives

4.2 FUNCTIONALITIES AND REQUIREMENTS

The general requirements and functionalities for the mission, and the different areas involved, are the following:

Functionalities	Requirements	Areas Involved
Stages	- The satellite shall be launched into Space. - The satellite shall be released by the mothership into the target orbit. - The entire mission shall have a low cost.	P-MA, P-PS, P-EPS
Manouver	- The satellite shall have an own space propulsion system in order to reach the positions defined. - The satellite shall perform an insertion maneuver to transfer from Europa Clipper's trajectory on its own orbit.	P-MA, P-PS, P-ADCS
Detach of 24U CubeSat from Europa Clipper	- Maximum size shall be 24 Units. - The system shall have a detach mechanism. - The detach maneuver shall not damage the mothership.	P-STR, P-ODCS

Functionalities	Requirements	Areas Involved
Orbit around Europa	- The orbit shall be defined. - The duration of the mission shall be defined. - The power budget shall be defined.	P-MA, P-TCS, P-EPS, P-PS, P-ADCS, P-ODCS
Attitude changes	- The satellite shall be able to change its attitude and its position. - The satellite shall be able to withstand the perturbations. - The satellite shall be able to satisfy the pointing requirements.	P-PS, P-ADCS
Thermal control	The satellite shall be able to keep the s/c components within the operational limits.	P-TCS
Orbit's correction	System Propulsion and ADCS components shall be activated automatically to keep the orbit defined.	P-PS, P-ADCS
Enable operations system	Applicable systems shall be deployed and ready for the next stage.	P-MA, P-EPS, P-STR, P-TMTC, P-TCS
Measure and evidence collection	The satellite shall collect data from the payloads whereas is moving on the orbit.	P-INS, P-ADCS, P-OB DH
Data collection	The satellite shall have the memory capacity to support all the data collection.	P-OB DH, P-TMTC
Telecommunication	The satellite shall have an antenna and point to Europa Clipper to send the information to the ground segment in a certain period of time.	P-OB DH, P-EPS, P-TMTC
Data analysis	- Build an entire topographic map of Europa's surface. - Define potential water plumes locations and chemical composition. - Define the chemical composition of Europa's atmosphere.	P-MA, P-OB DH
End of life phase	- Verify the possibility of a docking with Europa Clipper at the end of the mission. - Satisfy the planetary protection requirements for categories III and IV.	P-MA, P-PS, P-ADCS

Table 4.2 – Mission functionalities, requirements and areas involved

The areas involved are the following:

P-MA	P-TCS	P-EPS	P-STR	P-PS	P-ADCS	P-ODCS	P-TMTC	P-OB DH	P-INS
Mission Analysis	Thermal Control s\s	Electric Power s\s	Structure	Propulsion s\s	Attitude Determination & Control s\s	Orbital Determination & Control s\s	Telecommunication s\s	On Board Data Handling s\s	Instruments

Table 4.3 – Mission areas involved

5. MASPEX

5.1 INTRODUCTION

The MASPEX (MAss SPectrometer for Planetary Exploration) is a mass spectrometer specifically design for the unambiguous determination of volatile isotopes of methane, water, ammonia, carbon monoxide, molecular nitrogen, carbon dioxide, and low order (C2, C3, and C4) organic compounds in complex mixtures.

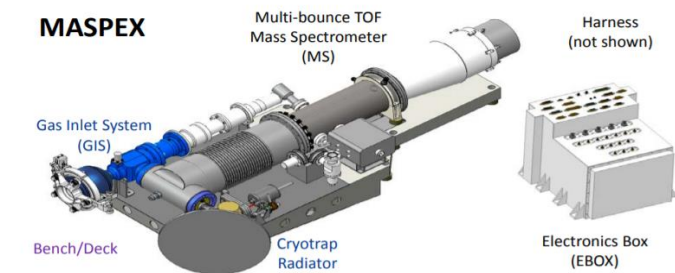


Figure 1 CAD drawing of the MASPEX Europa Engineering Model for the Clipper mission.

Figure 5.1 - MASPEX components [3]

This system is currently under development and it will be implemented on Europa Clipper, the Mission Carrier. It is capable to operate in a heavy radiation environment, with 540 rad per day, which is 115,2 krad for a complete scientific mission, and it is suitable for Category 3 Planetary Protection since it can be baked to 300°C against biological contamination.

It is composed by a GIS (Gas Inlet System), a multibounce time-of-flight mass spectrometer (MBTOF) and an EBOX (Electronics Box), as shown in *Figure 5.1*.

The GIS and the MBTOF weights 8 kg and the Ebox 2 kg, for a total of 10 kg for the whole instrument. It requires 6.4 W [4] in order to operate and it occupies 8U.

5.2 FUNCTIONAL OVERVIEW

A sample gas enters the system through the Gas Inlet System (GIS), it gets electron ionized and packets of ions are extracted into the MBTOF: here they will travel between two reflectrons and then they start bouncing. On each bounce, ions disperse in time according to their mass.

Once a particular number of bounces have occurred, ions will strike a micro-channel plate (MCP) electron multiplier. This device will emit a signal that will pass through the EBOX in order to record the ToF spectrum, as shown in *Figure 5.2*, and determine the ions' mass. The system is then capable of identifying each molecule by knowing the mass of the ions.

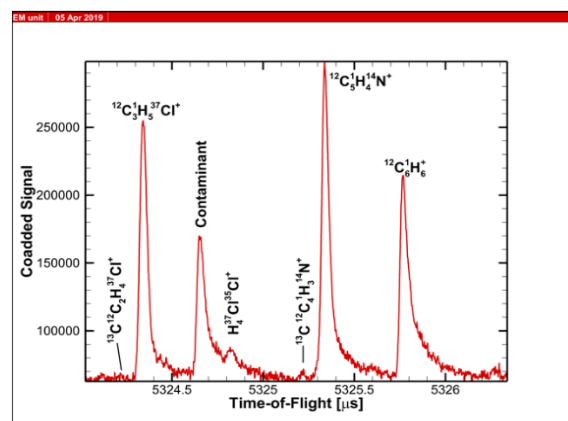


Figure 5.2 – Example of ToF spectrum [3]

The instrument can measure from 1-1500 u/z in a single spectrum and the longest collectable spectrum is 46.1 microseconds.

The system will produce 57 kbps [4] of data science and for the whole mission it is expected to produce 131.3 Gbyte.

5.3 OPERATIONAL METHOD

The instrument is initially set with customizable Science Operation Methods (SOMs). They are programmed before launch and can be modified by ground commands.

A SOM comprises a sample source and one or more regions of interest to span the required mass range. For each ROI there is an appropriate resolution and integration time in order to acquire the required analyte.

After orbit insertion over Europa, the system will run in Calibration Mode. Then, for the entire mission duration, SOM-A and SOM-T will run continuously.

In *Table 5.1* are reported the SOMs that will be implemented in the instrument and in *Table 5.2* are reported in detail the characteristics of SOM-A.

<i>Mode</i>	<i>Description</i>	<i>Characteristics</i>
SOM-A	Ambient mode. Specific chemical composition analysis with spatial resolution	23 s of varying length co-added spectra in each of 4 ROIs, mass range 2-32 u, $m/\Delta m$ 6,948.
SOM-T	Cryotrapping mode. Prior to mode C, parallel to mode A.	Traps sample on adsorber at ~65K; Sample density determines trap time
SOM-C	Cryotrapped sample survey. Enables method updates to suit environment. Comprehensive mass survey, ppm mixing fraction species.	0.5 s co-added spectra in each of 34 ROIs, mass range 2-200 u, $m/\Delta m$ ~12,000, total time 17 s.
	Cryotrapped sample analysis. Targeted ROI method minimizing analysis time and data volume.	2028 s of varying length co-added spectra in each of 13 ROIs, mass range 16-114 u, $m/\Delta m$ up to ~23,822.

Table 5.1 - Science Operation Methods implemented on MASPEX [5]

ROI	Integration time (s)	Analyte	Mass (u)	Resolution (m/ Δ m)
1	16.71	HCN	27.01	3,690
		C ₂ H ₄	28.03	6,948
		C ₂ H ₆	30.05	6,948
2	0.55	O ₂	31.99	3,042
3	1.36	CH ₄	16.03	1,464
		H ₂ O	18.01	91
4	4.37	H ₂	2.02	31
		HD	3.02	37

Note: m/ Δ m uses 10% valley definition.

Table 5.2 - Characteristics of SOM-A [5]

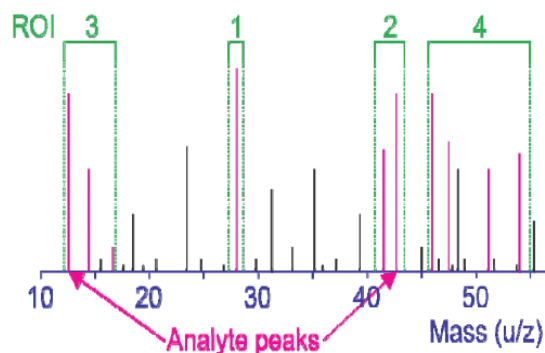


Figure 5.3 - Example of TOF produced during SOM-A highlighting the ROI [5]

5.3 REQUIREMENTS

REQ. ID	IMPORTANCE	REQUIREMENTS
RHAD-INS-010	M	The instrument shall run in calibration mode after correct orbit insertion.
RHAD-INS-020	M	When the system is acquiring gas samples, the propulsion system shall not be used in order to not contaminate the sample gases and invalidate the measurements.
RHAD-INS-030	M	GIS pointing offset design shall satisfy spectrometry requirements, so that the incoming vapour particles mass flow rate meets the required threshold defined by the manufacturer for an effective sampling.

Table 5.3 - MASPEX requirements

6. CHAMALEON IMAGER

6.1 INTRODUCTION

The Chameleon is a compact CubeSat imager providing high resolution imaging in different configurations.

The Hyperspectral linescan setup may offer relevantly interesting remote sensing of biogenic indicators over Europa soil.



Figure 6.1 - Chameleon Imager [6]

- Trying to assess the habitability of Jupiter's moon, key spectral absorption features have been detected in both the icy and the non-icy Europa materials, so the determination of planetary surface composition from remote infrared spectroscopy is relevant within NIR and SWIR spectral bands, to distinguish spectral features of the compounds of interest [10].
- This camera shall also perform surface mapping within VIS spectrum so to study its features, how these relate to each other and to sub-surface structures. Moreover, it will look for potential plumes venting vapors into space and signs of geologic activity [7].

These elements together will be essential for subsequently defining possible landing and sample collection spots for future missions.

6.2 FUNCTIONAL OVERVIEW

Concerning the spectral analysis, the payload is capable of recording 148 bands in the 0.8-2.5 μm region (NIR) which is essential for quantifying sulfuric acid hydrates and bulk surface composition [8]. Extending the infrared spectra up to 5 μm , the presence and distinction of amino acids (leucine, isoleucine and their enantiomers) [9], CO_2 , H_2O , H_2O_2 , SO_2 , hydrated salts and more organics shall be feasible [10].

The pan-sharpening process to enhance the final resolution is comprised, in order to obtain competitive results with respect to Europa Clipper's imaging system payload (EIS).

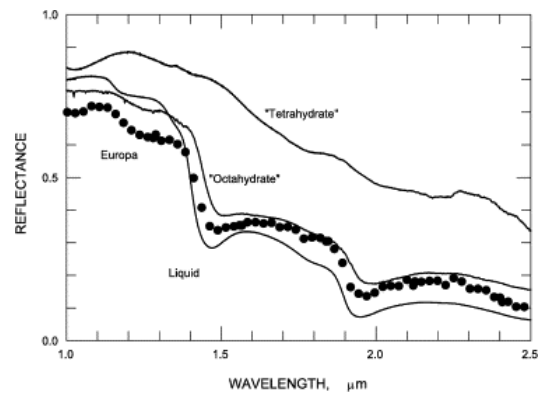


Figure 6.2 - Diffuse-reflectance spectra of solid and liquid sulfuric acid hydrates and Europa [11]

	CHAMELEON IMG	EIS: NAC / WAC
Resolution (Push-broom observation)	3333p PAN 1052p HS	8000p (<5%) / ≤80p (>95%) NAC 1375p WAC
FOV	2.3°	2.3° NAC / 48° WAC

Table 6.1 - Imaging Specifics of Chameleon Imager in comparison with EIS Narrow Angle Camera (NAC) and Wide-Angle Camera

6.3 OPERATIONAL METHOD

The mission design shall guarantee constant altitude so to perform 100% surface mapping at constant Ground Sample Distance. The resulting GSD is competitive with EIS.

CHAMELEON IMG @100km	EIS @50km fly-by	
1.2 m/pixel	0.5 m/pixel	32 m/pixel
100%	<5% NAC 1	WAC
	50 m/pixel	
	>95% NAC 2	

Table 6.2 - Chameleon Imager GSD in comparison with EIS [6-7]

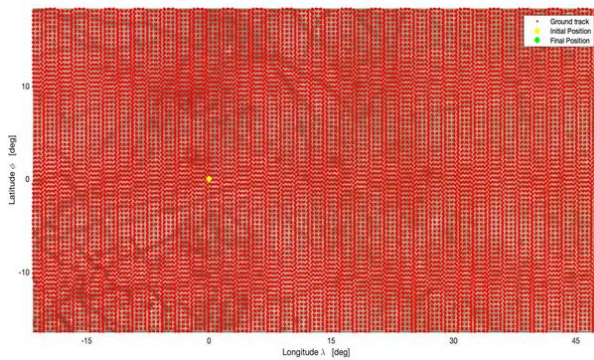


Figure 6.3 - The surface is fully accessible at max resolution of 1.2m/p by Chameleon Imager

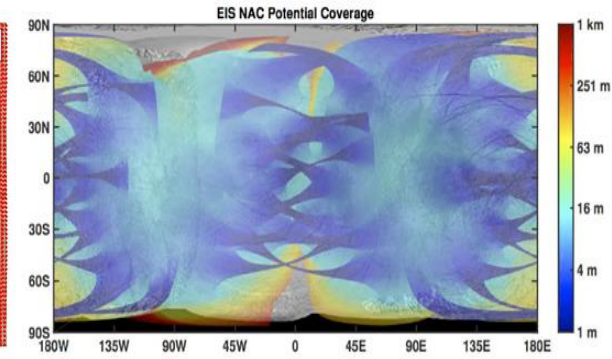


Figure 6.4 - Areas of Europa's surface accessible by Europa Clipper Imaging System, Narrow Angle Camera, color-coded by pixel scale (mapping of >95% of Europa at ≤50m/pixel) [7]

Rhadamanthus and Europa Clipper's imaging missions are quite complementary. The first one shall provide an optimal and continuous recording, involving a higher data volume critical handling requirement and critical structural decay due to high radiative environment (order of 10^2 krad per the expected scientific mission duration). This system isn't capable of providing downlink to Earth station also due to antennas limitations, hence it's expected to send telemetry and scientific data to the mothership first. In fact, EC's path is designed to grant higher longevity, orbiting far away from the hazardous radiative environment of Europa, therefore it offers more time devoted to data transfer.

In order to guarantee constant frame rate and resolution during the imaging, the bitrate requires to be modulated, so to avoid pixelation, but still aiming to decrease the file size as much as possible.

CHAMELEON IMAGER Data Rate REQs	
Spatial resolution (GSD)	1.2 m/pixel @ 100km altitude
Rec Frame rate	15 fps
Tuning range bitrate of rec	15-50 Mbps
Recorded Data Volume	<115.182 TByte

Table 6.3 - Example of data rate requirement considering one complete scientific mission of 213.3 days without repetitions

REQ. ID	IMPORTANCE	REQUIREMENTS
RHAD-INS-040	M	Pointing design shall include a pivoting kinematics to stabilize the video frames during acquisition.
RHAD-INS-050	M	Altitude shall be maintained constant
RHAD-INS-060	M	The software shall detect altitude irregularities and evaluate to remap the same area
RHAD-INS-070	M	The software shall detect frame losses and evaluate to remap the same area
RHAD-INS-080	M	Compression algorithms shall guarantee downlink is feasible, with respect to antennas limitations

Table 6.4 - Chameleon Imager requirements

7. MISSION ARCHITECTURE

7.1 CUBESAT - 2024

The architecture is based on an only powerful CubeSat which shall be constrained to a maximum of 2 scientific payloads and 24 Units, not including the solar panels. The tasks are mapping Europa's surface with HD sensors, sample Europa's atmosphere and water plumes and transmit the data to the ground station.

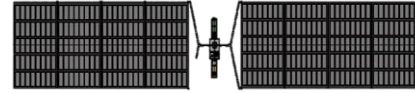


Figure 7.1 – Rhadamanthus spacecraft

7.2 MOTHERSHIP - 2024

The CS shall be compatible with a nominal launch on Europa Clipper aboard the SpaceX's Falcon Heavy from Kennedy Space Center, USA in 2024. Nevertheless, the spacecraft shall be adaptable with a backup on Vulcan Centaur from KSC.



Figure 7.2 – Europa Clipper logo [13]

7.3 MARS AND EARTH GRAVITY ASSIST - 2025/2026

Europa Clipper shall perform two flybys around Mars and Earth during the interplanetary trajectory to reach Jupiter, in order to achieve the power needed. The path shall follow a flyby of Mars in 2025 and a flyby of Earth in 2026, reaching Jupiter system in 2030.

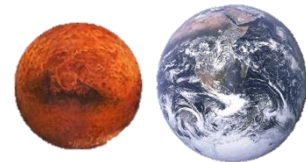


Figure 7.3 – Mars and Earth

7.4 DETACHMENT AND READINESS - 2030

The system will reach Jupiter system in April 2030, at the arrival the probe shall perform detachment from the second stage of Falcon Heavy and correct its attitude and trajectory to start orbiting around Jupiter and performing multiple flybys around Europa. In addition, the CubeSat shall detach from Europa Clipper during the first suitable flyby around Europa and it's strictly constrained to not interface or damage the mothership.

Subsequently the CubeSat shall deploy parts needed and perform a quality check of software and hardware and then correct attitude, velocity and trajectory, in order to start orbiting around Europa.



Figure 7.4 – Europa Clipper [14] and Rhadamanthus

7.5 ORBIT THROUGH EUROPA - 2030

The CubeSat shall be inserted into a circular and polar orbit, with a 100km altitude performing at least 2452 orbits around Europa. The CubeSat shall survive at least 7 months in its orbit around Europa, to successfully accomplish the mission

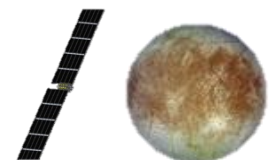


Figure 7.5 – Europa and Rhadamanthus

7.6 END OF LIFE - 2030

For the end-of-life phase of the mission, the possibility of a docking with Europa Clipper shall be evaluated, in order to avoid the crash of the CubeSat on Europa's surface. In any case, the requirements regarding the planetary protection for categories III/IV, to which Europa belongs, shall be satisfied in the case of an impact on Europa.

8. MISSION ANALYSIS

8.1 INTRODUCTION AND REQUIREMENTS

In this chapter are presented the Mission Analysis results for the Rhadamanthus mission, considering the orbit insertion from Europa Clipper's orbit and the trajectory of the spacecraft needed to accomplish the objectives of the mission.

The principal mission analysis requirements are reported in *Table 8.1*:

REQ. ID	IMPORTANCE	REQUIREMENTS
RHAD-MA-010	M	The 24U CubeSat shall be compatible with a nominal launch on Europa Clipper probe, aboard the SpaceX's Falcon Heavy, from Kennedy Space Center.
RHAD-MA-020	M	The spacecraft shall be compatible with a backup launch on Vulcan Centaur from KSC.
RHAD-MA-030	M	The CubeSat shall detach from EC during the first suitable flyby around Europa
RHAD-MA-040	M	The CubeSat shall perform an insertion maneuver to be inserted in the desired orbit
RHAD-MA-050	M	The insertion maneuver shall be compatible with the propulsive subsystem constraints in terms of Δv .
RHAD-MA-060	M	The spacecraft shall perform at least 2452 orbits around Europa.
RHAD-MA-070	M	The possibility of a docking with Europa Clipper at the end of the mission shall be evaluated.
RHAD-MA-080	M	The requirements for the planetary protection for category 3 and 4 shall be satisfied in case of crash on Europa.

Table 8.1 – Mission Analysis requirements

8.2 EUROPA CLIPPER'S TRAJECTORY AND TIME SCHEDULE

Europa Clipper is an interplanetary mission in development by NASA, employing as launcher the SpaceX's Falcon Heavy. The launch date from the Earth is expected for October 2024 during a 21-day launch window. The spacecraft will use gravity assists from Mars in February 2025 and Earth in December 2026.

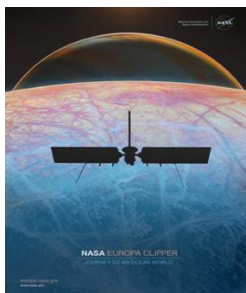


Figure 8.2 - Europa Clipper wallpaper, NASA JPL [15]

The first orbital insertion in Jovian system is scheduled for April 2030. After the arrival in Jupiter system, Europa Clipper will start to perform multiple flybys around Europa, with different altitudes, inclinations and eccentricity. In one of the most suitable flybys (in terms of eccentricity and inclination), Europa Clipper will release the RADHAMANTHUS spacecraft into the Europa's sphere of influence. The insertion maneuver shall be coherent with the mission objectives.



Figure 8.1 -Europa Clipper logo, NASA JPL [13]

8.3 ORBIT'S ANALYSIS

The chosen orbit for the mission is a circular and equatorial orbit, with an altitude of 100 km, chosen considering the field of view of Chameleon Imager and the need to have a repeating ground track after 2452 orbits without overlapping, minimizing the lifetime of the mission.

The orbit is circular in order to have a constant distance from Europa's surface, allowing the chameleon imager to obtain a better resolution in imaging.

The orbit has an inclination of 90° to cover the entire surface of Europa, poles included.

The orbit's characteristics are reported in the *Table 8.2*, and the trajectory is represented in the *Figure 8.3*.

Eccentricity	$e = 0$
Inclination	$i = 0^\circ$
Radius of Europa	$R_E = 1560.8 \text{ km}$
Altitude	$h = 100 \text{ km}$
Semi-major axis	$a = 1660.8 \text{ km}$
Period	$T = 7515 \text{ s}$

Table 8.2 – Orbit's characteristics

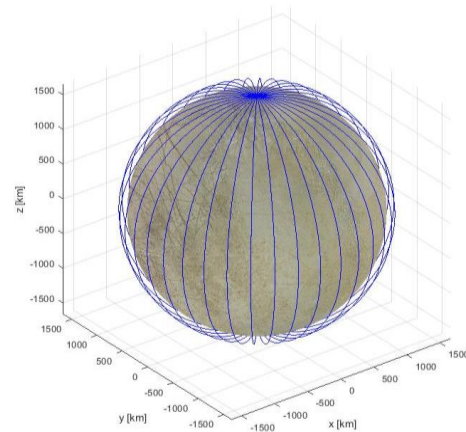


Figure 8.3 – Trajectory of the spacecraft around Europa

8.4 GROUND TRACK OF EUROPA'S SURFACE

The chosen orbit is able to cover the entire Europa's surface without any significant overlap of the path. To cover the entire surface, it's necessary to perform at least 2452 revolution, considering the area covered by the chameleon imager at the altitude chose, and the time employed to perform one revolution and the rotation period of Europa.

For 100 revolutions, in *Figure 8.4* is reported the ground track of the spacecraft's orbit. The ground track for 2452 period cover the entire surface, so a zoomed part is reported in *Figure 8.5*:

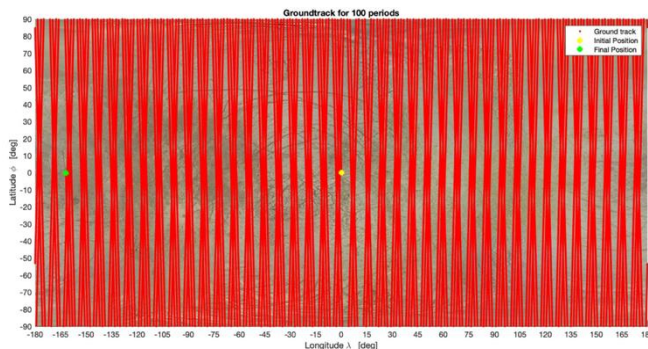


Figure 8.4 – Ground track for 100 periods

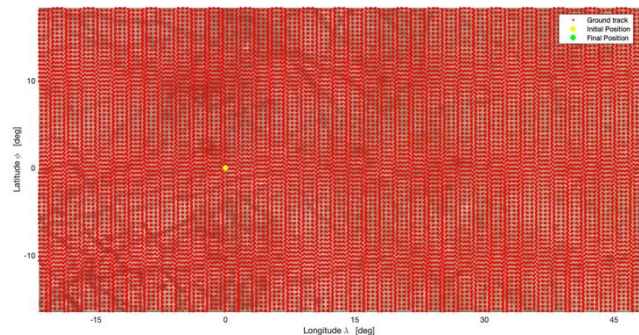


Figure 8.5 – Zoomed ground track for 2452 periods

8.5 ORBIT INSERTION

Europa Clipper performs multiple flybys around Europa, with different inclination, eccentricity and perigee altitude.

The CubeSat will be released at the pericenter of a hyperbolic arc when the first suitable fly-by of Europa will take place at an altitude equal to 100 km from Europa's surface. Neither the inclination of the hyperbola with respect to the equator of Europa and the eccentricity of the incoming arc are exactly known. With the "graphical" data provided by NASA (Figure 8.6), it's possible to assume a case of very high inclination with respect to the equatorial plane, when Europa clipper is closer to Europa, and also a quite low eccentricity of the incoming arc of hyperbola during flyby.

Assuming a pericenter altitude of 100 km from Europa's surface, an inclination of 78° and an eccentricity of 1.01, the Δv required to insert the spacecraft in a circular and polar orbit, is equal to $\Delta v = 0.685 \text{ km/s}$.

The maneuver performed to insert the spacecraft in the desired orbit consists in a plane change maneuver with a deceleration, from the periapsis velocity of the incoming hyperbola to the velocity on a circular orbit. The shape change and inclination change maneuver are performed together to minimize the Δv and save propellant.

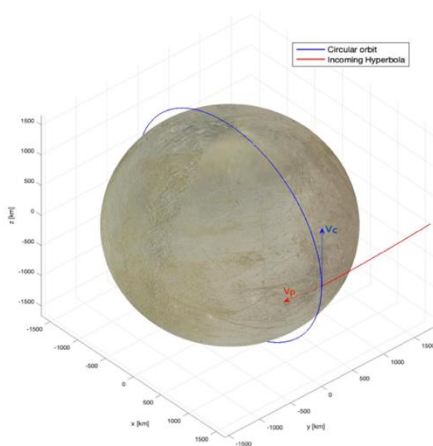


Figure 8.7 – Orbit insertion

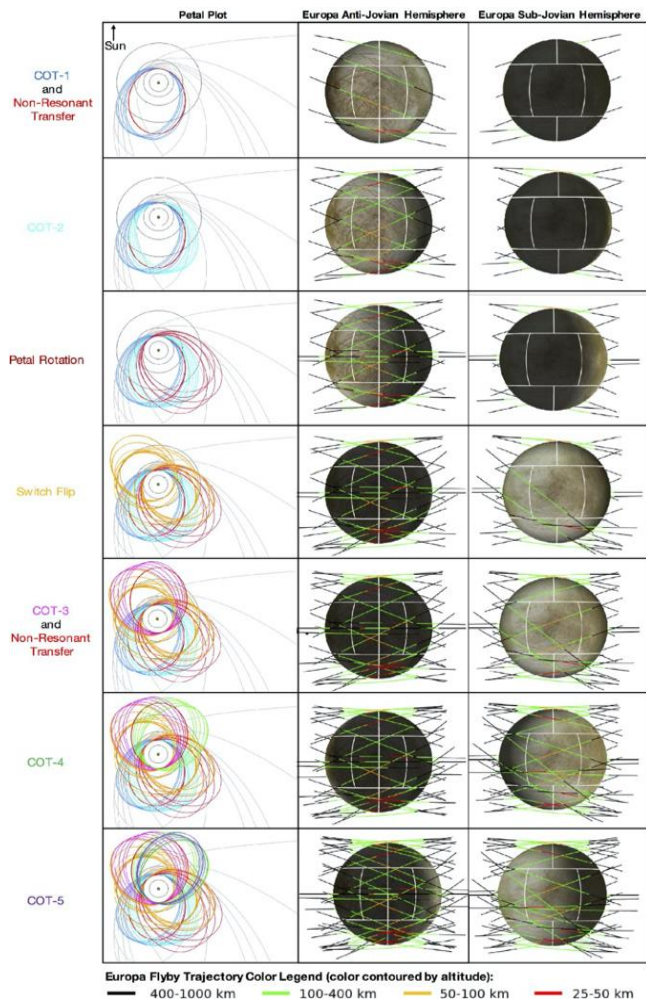


Figure 8.6 – Fly-by Maneuvers of Europa Clipper, NASA JPL [16]

In Figure 8.7 is reported the insertion maneuver, from the pericenter of the hyperbola to the circular orbit. In the figure, the red orbit represents the hyperbolic flyby while the blue orbit represents the spacecraft's trajectory. The relative inclination between the two orbits is exaggerated to stress the difference between the orbits. Considering:

- $i_{hyp} = 78^\circ$;
- $i_{circ} = 90^\circ$;
- $e_{hyp} = 1.01$;

The result obtained is: $\Delta v = 0.685 \text{ km/s}$

9. ATTITUDE DYNAMIC AND CONTROL SYSTEM

9.1 INTRODUCTION AND REQUIREMENTS

The Attitude Determination and Control Subsystem (ADCS) has the objective to determine the CubeSat attitude at any instant during the mission life, and to control the satellite in order to guarantee the success of the mission goals.

Additional duties of the ADCS are related to other subsystems, for example with Electrical Power Subsystem (EPS) for which the solar arrays must point as longer as possible towards the Sun in order to charge secondary batteries on board. Another duty is linked with the Telemetry Tracking and Telecommand Subsystem (TTMTCS), to make the antenna point the desired direction and guarantee the data transmission to Europa Clipper.

The set of requirements imposed to the ADCS is listed in *Table 9.1*.

REQ. ID	IMPORTANCE	REQUIREMENTS
RHAD-ADCS-010	M	The ADCS shall provide a robust control scheme.
RHAD-ADCS-020	M	The ADCS shall provide the success of the de-tumbling phase.
RHAD-ADCS-030	M	The ADCS shall be able to provide slew maneuvers if needed.
RHAD-ADCS-040	M	The ADCS shall guarantee the correct tracking of Europa.
RHAD-ADCS-050	M	The ADCS shall be able to reject external disturbance torques.
RHAD-ADCS-060	M	The ADCS shall be able to de-tumble the spacecraft within 180 s after the releasing.
RHAD-ADCS-070	M	The CubeSat shall be released with angular velocities along its body axes less than 22°/s.
RHAD-ADCS-080	M	The ADCS shall guarantee a Nadir pointing for the imager with an accuracy of 0.2°.
RHAD-ADCS-090	M	The ADCS shall keep solar arrays pointed towards the Sun with an accuracy of at least 10°.
RHAD-ADCS-100	M	The ADCS shall keep the antenna pointed in the outward direction of the orbit with less than 20° accuracy.
RHAD-ADCS-110	M	The ADCS shall determine the attitude with at least one order of magnitude higher with respect to the pointing accuracy, along each axis.

Table 9.1 - ADCS requirements

9.2 DESIGN PATH

In order to preliminary size the entire subsystem, the main control modes have been identified, to then determine pointing direction for specific instruments with a certain accuracy depending on the mission phase. Then, on the basis of instruments' needs, a pointing budget has been filled to evaluate the accuracy needed for the attitude determination. Finally, the attitude control architecture has been chosen.

9.2.1 CONTROL MODES

Analyzing the mission profile and the mission goals, three main control modes have been identified.

De-tumbling mode This control mode begins once the CubeSat is released by Europa Clipper at the perigee of a hyperbolic arc at an altitude of 100km. It consists in de-spinning the satellite as soon as possible to perform the desired insertion maneuver; it's also the moment in which the CubeSat acquires its initial attitude and the mode can be used to recover from emergencies.

This mode shall be performed within 180s to guarantee the success of the insertion maneuver; for this reason the CubeSat shall be released such that to have angular velocities less than $22^\circ/\text{s}$ along its body axes.

Normal on-board mode This mode is applied for the majority of the mission lifetime. Once the CubeSat is inserted into the nominal orbit this mode starts. The main drivers are in this case the different pointing directions and accuracies of the instruments; the Chameleon imager shall point the Nadir direction, solar arrays shall point the Sun and the antenna shall point the outward direction of the orbit. This mode must also guarantee the external torques rejection. For all these reasons the requirement from this mode drives the sizing process.

Slew mode This control mode is thought whenever a re-alignment maneuver is needed, starting from the correct alignment of the thruster before performing the insertion maneuver, after de-tumbling, but also during the nominal orbit, to recover from disturbances or permit a better functioning of payloads. A reason could be represented by the need of the TMTCS to transmit data to Europa Clipper, which is not at that instant in the correct antenna direction.

9.2.2 POINTING BUDGET AND ARCHITECTURE

Exploiting the previous information and the needs of some instruments, the most restrictive value for what regards the accuracy is given by the Chameleon Imager, in particular with a pointing accuracy of 0.2° , against 10° of solar panels and 20° of the antenna.

Starting from this value the attitude determination shall be at least one order of magnitude more accurate.

Since the need of multiple pointing requirements, so different direction pointing, and the need of high pointing accuracy (0.2°), the 3-axis control has been selected as the most suitable control architecture. This choice is also coherent with the need of very large solar panels (which will be presented in the EPS chapter) with respect to the CubeSat's structure.

9.3 EXTERNAL DISTURBANCES

The CubeSat, during its mission life, will be inevitably subjected to several external disturbances, which can produce a net torque along the spacecraft and modify its attitude. The quantification of these effects is relevant for the sizing of the subsystem, in particular to determine the level of authority required for the actuators.

As main sources of external disturbances gravity gradient, aerodynamic drag and solar radiation pressure (SRP) have been identified. Since the lack of reliable data and due to the complicated mechanism of the Jovian induced magnetic field on Europa, also variable in time, the magnetic field has not been considered in the study. Also, external sources due to for

example the rotation of solar arrays towards the Sun have not been considered; it has been assumed that the latter affects less the system with respect the disturbances presented before.

In *Figure 9.1*, *Figure 9.2* and *Figure 9.3* are shown the torques along each body axis as consequences of respectively gravity gradient, SRP and aerodynamic drag, evaluated in one period of the CubeSat.

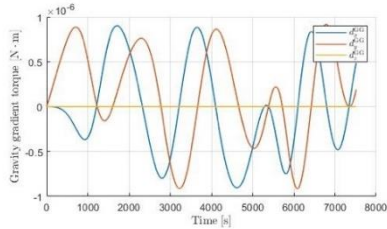


Figure 9.1 - Gravity gradient torque

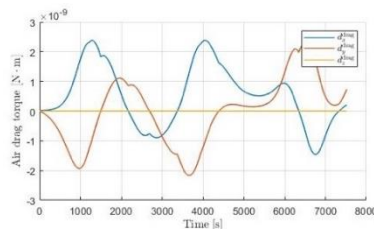


Figure 9.2 - Aerodynamic drag torque

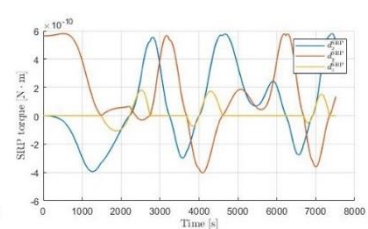


Figure 9.3 - SRP torque

While in *Figure 9.4*, it is shown the evolution of the overall torque along each axis, in one period.

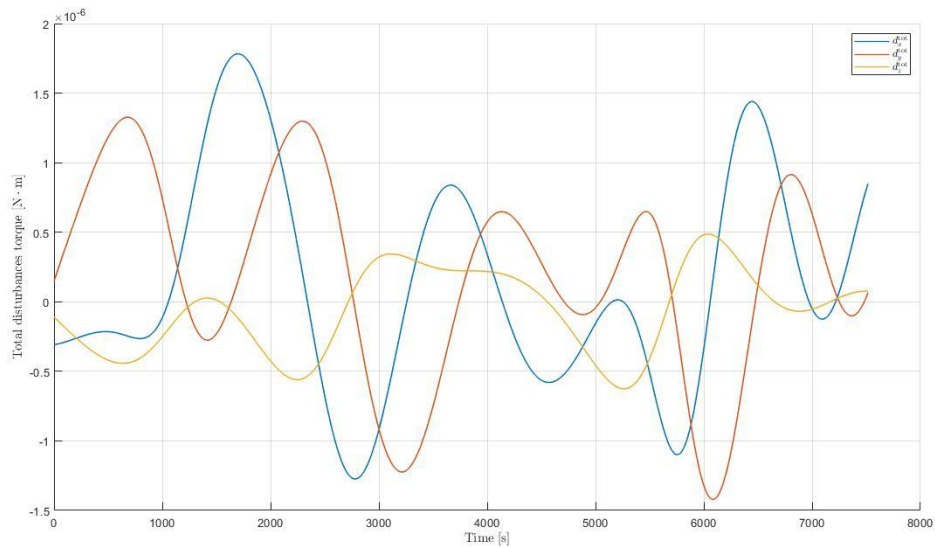


Figure 9.4 - Overall external torque

Looking at the previous images, it is possible to notice how the aerodynamic drag torque is not so relevant due to the weak atmosphere of Europa; the SRP torque provides low values and this is expected due to the high distance of Europa from the Sun, around 5.2 AU. In the images eclipse phases are not considered, also this point has to be taken into account, which leads to null SRP torque during eclipses.

The overall external torque results in the order of $10^{-6} Nm$.

9.4 HARDWARE

9.4.1 SENSORS

The choice of the determination hardware is relevant to guarantee an accurate and rapid response of the system to reconstruct the current attitude and to provide a precise response to control commands.

Exploiting the information presented in 9.2, Sun sensors and stellar gyros are used to reconstruct the state of the satellite, while 3 MEMS gyroscopes mounted along each axis are employed to retrieve the angular rate of the CubeSat. In particular, a solution with three 0.5° accurate Sun sensors and three 0.05° accurate stellar gyros has been adopted. The choice permits to retrieve measurements along each axis of the spacecraft, it guarantees redundancy in case of failure of one or more sensors and also it provides the correct functioning during eclipses. Since the CubeSat is exposed to eclipse, both of Europa and Jupiter, for a wide duration of the mission, stellar gyros are very relevant.

All these sensors are mounted on a single PC board [17], whose properties are reported in *Table 9.2*.

Dimensions	96 x 96 x 15 mm	Mass	0.450 kg
Power required	3.0 W	Power supply	5V and 3.3V
Operational Temperature	-25°C to +50°C	Radiation tolerance	10 krad

Table 9.2 - ADC board properties

Considering as reference value of the radiation dose equal to 540 rad/day [18] (value at Europa's surface) and considering the mission lifetime presented in Chapter 8, it results obvious that the previous instrumentation shall have an adequate shielding to prevent them from degradation.

9.4.2 ACTUATORS

The selection of the actuators has been derived from the previous considerations about the external disturbances and pointing requirements. The choice is between Reaction Wheels (RW) and Control Moment Gyroscopes (CMG), but the latter are more suitable for high specific torques and imply more inert mass.

For these reasons, in order to accomplish the desired functionalities, a set of four Reaction Wheels has been chosen. These wheels are arranged in a tetrahedral configuration, as shown in *Figure 9.5*, to guarantee robustness and reliability also in case of failure of one wheel.

The RWs' properties [19] are resumed in *Table 9.3*.



Figure 9.5 - RW configuration

Dimensions (per wheel)	70 x 70 x 48 mm
Mass (per wheel)	1.0 kg
Max. torque (per wheel)	0.02 Nm
Max. momentum (per wheel)	0.2 Nms
Power (steady state, 4000 rpm)	3.0 W
Operational temperature	-20°C to +70°C
Radiation tolerance	20 krad
Configuration	Tetrahedral

Table 9.3 - RWs' properties

One important aspect that regards the use of reaction wheels is the saturation level of the instrument; once it reaches its maximum momentum storage it must be de-saturated. Since the ADCS doesn't contain additional actuators such as magnetorquers or cold gas thrusters, which could de-saturate RWs, the latter shall be maintained below their 50% of maximum momentum storage, to have a margin of operability. Starting from this consideration, the maximum slew rate admissible by the system shall be evaluated. As for the case of sensors, also RWs require an adequate shielding to protect them from radiations.

9.5 SOFTWARE

The feasibility and effectiveness of the architecture presented before have been proved through different simulations developed in MathWork's Simulink.

In this paragraph the main implementations for what regards the attitude and control actions will be shown.

9.5.1 ATTITUDE DETERMINATION ALGORITHM

In order to determine the current attitude of the spacecraft, the vector measurements coming from stellar gyros and Sun sensors, with their respective accuracy, are exploited. In particular, they are employed to minimize the weighted cost function J belonging to Whaba's problem:

$$J = \frac{1}{2} \sum_{i=1}^N \alpha_i \|\underline{v}_{Bi} - A_{B/N} \underline{v}_{Ni}\|, \text{ where } \alpha_i \text{ such that: } \sum_i \alpha_i = 1$$

Both the measurements in the inertial frame (denoted with "n"), obtained from the propagation of the orbit, and in the body frame (denoted with "b"), obtained from the sensors, are available; to minimize the previous weighted function the Singular Value Decomposition (SVD) has been used [20].

In this way, the procedure to compute the attitude direction cosines matrix from on-board sensors is the following:

$$B = \sum_{i=1}^N \alpha_i \underline{v}_{Bi} \underline{v}_{Ni}^T = (\text{through SVD}) = USV^T$$

$$A_{B/N}^{sensor} = U M V^T \quad \text{where } M = \text{diag}(1, 1, \det(U) \cdot \det(V))$$

9.5.2 FILTERING ACTIONS

The attitude recovered by sensors will be inevitably affected by noise; in order to avoid this problem and to obtain a clear estimated attitude, a state observer has been implemented as follows:

$$\dot{A}_{B/N}^{est} = -[\underline{\omega}^{\wedge}] A_{B/N}^{est} + \alpha_{obs1} (A_{B/N}^{est} - A_{B/N}^{sensor})$$

The same procedure has been adopted for the angular velocity estimated by the gyroscopes:

$$I \dot{\underline{\omega}}_{est} = I \underline{\omega}_{est} \times \underline{\omega}_{est} + \underline{d} + \underline{u} + \alpha_{obs2} (\underline{\omega}_{est} - \underline{\omega}_{gyro})$$

During the simulation, both ARW (Angular Random Walk) and RRW (Bias Random Walk) have been handled as Gaussian noise, with zero mean value, and standard deviation according to the data provided by the instruments.

The angular velocity measured by the gyroscope will be affected by the presence of both these two noises.

$$ARW: \underline{n} = \sigma_n \underline{\xi}_n \quad RRW: \underline{b} = \sigma_b \underline{\xi}_b \quad \underline{\omega}_{gyro} = \underline{\omega} + \underline{n} + \underline{b}$$

Alternatively to state observers, low pass filters could be implemented in order to filter undesired noise, but in this way a phase delay would occur, with a resulting control action delay.

For these reasons state observers were preferred during the simulation.

9.5.3 CONTROL LOGICS

De-tumbling The control logic adopted during the de-tumbling phase is a simple and continuous control law, implemented as follows:

$$\underline{u}_c = -k\underline{\omega}$$

The torque and the control action provided by the reaction wheels can be computed as:

$$\underline{\dot{h}}_r = -A_{rw}^* (\underline{u}_c + \underline{\omega} \times A_{rw} \underline{h}_r) \quad \underline{u}_{rw} = -A_{rw} \underline{\dot{h}}_r - \underline{\omega} \times A_{rw} \underline{h}_r$$

Several simulations have been carried out, each one with random initial conditions and different values for initial angular velocities after the release of the CubeSat.

Here, in *Figure 9.6*, an example of the described control action is shown; the real control torque corresponds to the ideal one, showing the robustness of the algorithm.

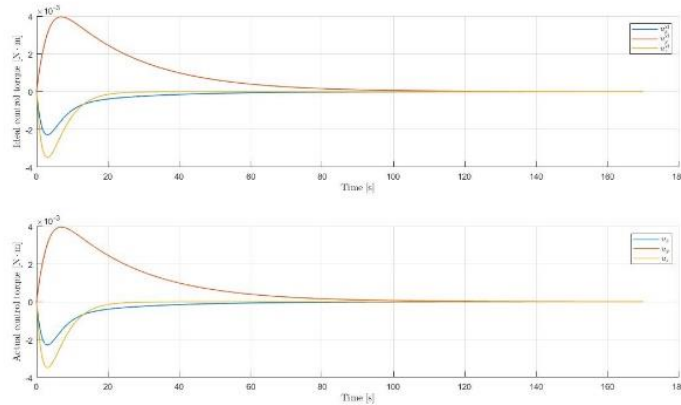


Figure 9.6 - Control action during de-tumbling

For a more accurate analysis and more effective results, a Monte Carlo simulation is well suited.

Slew maneuvers The control logic adopted during this phase is a non-linear, feedback based control law, implemented as follows:

$$\underline{u}_c = -k_1 \underline{\omega} - k_2 (A^T - A)^v$$

The controller, in this case, can be treated as a Proportional-Derivative (PD) controller, where the derivative controller takes as input the angular rate of the CubeSat.

Europa pointing The tracking control is more complex than the slew control; it is exploited to track a time dependent attitude trajectory (in this case LVLH frame) and it requires precise

knowledge of the acceleration of the moving frame. The control logic, also non-linear and feedback based, is the following:

$$\underline{u}_c = -k_1 \underline{\omega}_e - k_2 (A_e^T - A_e)^V + \underline{\omega} \times I \underline{\omega} + I (A_e \dot{\underline{\omega}}_d - \underline{\omega}_e \times A_e \underline{\omega}_d)$$

The control action provided by RWs is the same as the one presented for de-tumbling phase. The results shown in Figure 9.7 and Figure 9.8 describe respectively the evolution of the control action and of the attitude estimation along one orbital period, once the CubeSat is already aligned with the moving frame.

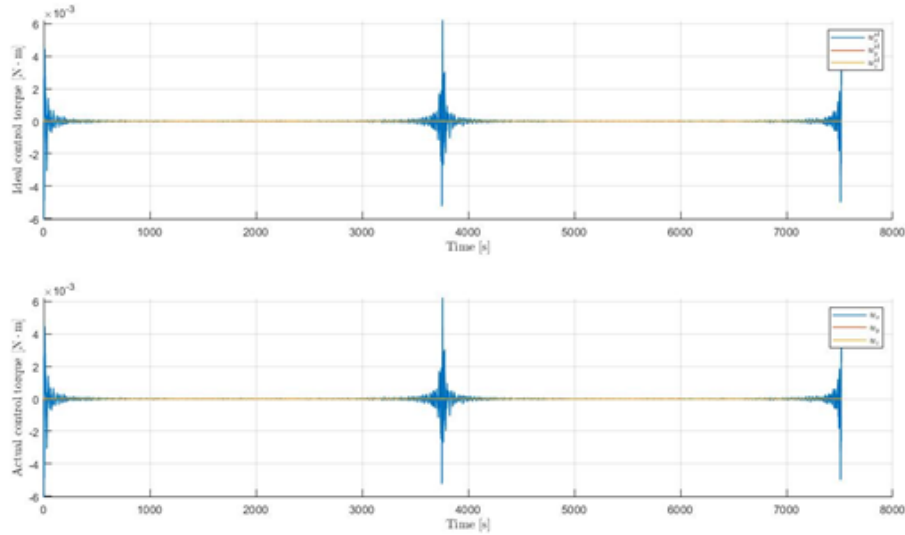


Figure 9.7 - Control action in one orbital period pointing Europa

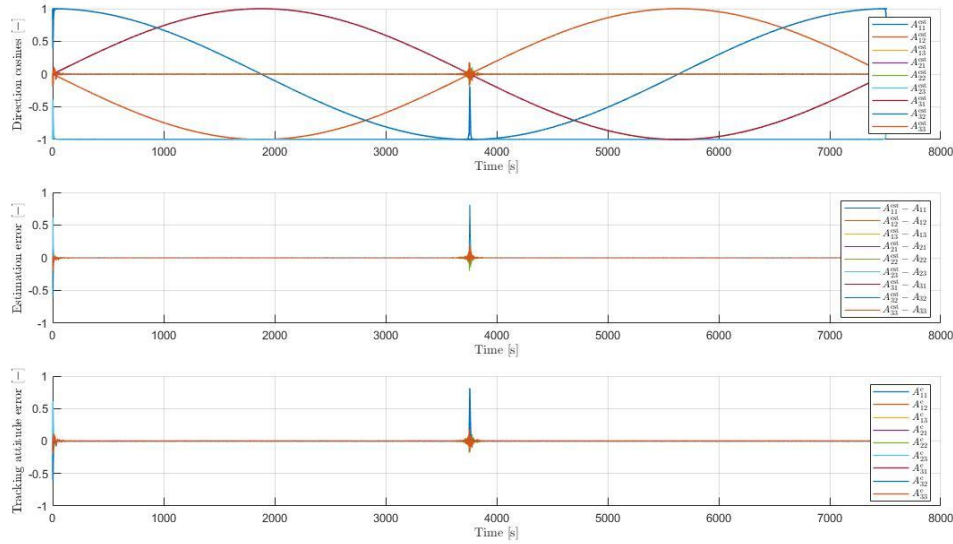


Figure 9.8 - Attitude evolution in one orbital period pointing Europa

Also in this case the adopted control logic guarantee robustness with respect to external disturbances; the attitude estimation error present during half of the orbit has to be accounted to singularities in the attitude model.

10. TELEMETRY, TRACKING AND TELECOMMAND SUBSYSTEM

10.1 INTRODUCTION AND REQUIREMENTS

The Telemetry, Tracking and Telecommand subsystem is in charge of sending the telemetry and data science of the mission and of receiving commands from ground stations.

As a first step the distance between Europa and Earth, at the estimated Europa Clipper arrival in the Jupiter system at 2/04/2030 was computed.

The distance is: $d = 810.672 \cdot 10^9 \text{ km} = 5.48 \text{ AU}$

Sizing the system with this link distance, in order to communicate directly with Earth, was a problem, because it is necessary to consider that most of CubeSat telecom components work with low power transmission and low gain. Considering also that dimensions are a constraint in the mission, this link would operate with low E_b/N_0 and low data rate.

Further analysis has led to the conclusion that Rhadamanthus mission will communicate with Europa Clipper, the Mission Carrier.

REQ.ID	IMPORTANCE	REQUIREMENTS
RHAD-TMTC-010	M	The TMTC, in data mode, shall produce a signal with a minimum $E_b/N_0 = 9 \text{ dB}$, in order to have a minimum $BER=10^{-5}$
RHAD-TMTC-020	M	The TMTC, in telemetry mode, shall produce a signal with a minimum $E_b/N_0= 10.5 \text{ dB}$, in order to have a minimum $BER = 10^{-6}$
RHAD-TMTC-030	M	The TMTC, in command mode, shall receive a signal with a minimum $E_b/N_0 = 11 \text{ dB}$, in order to have a minimum $BER = 10^{-7}$
RHAD-TMTC-040	M	EC shall transmit Rhadamanthus data.
RHAD-TMTC-050	M	In long-term planning shall be considered to send commands from Ground Stations to Rhadamanthus mission via Europa Clipper.
RHAD-TMTC-060	M	A beacon shall be implemented in the TMTC subsystem, so that in case of failure of the system it will send telemetry data.

Table 10.1 - Telecom s/s requirements

10.2 COMPONENTS

The parameters of the Europa Clipper High Gain Antenna (HGA) [21] are the following:

- X-Band uplink frequency: $f_{TX} = 7170$ MHz
- X-Band downlink frequency: $f_{RX} = 8424$ MHz
- $G_{TX} = 44.7$ dBic
- $G_{RX} = 46$ dBic

The characteristics of the EC antenna has led to the choice of the components of Rhadamanthus telecom subsystem.



Figure 10.1 – Europa Clipper HGA prototype [21]

The subsystem is composed by one transceiver, two antennas, one power amplifier and a beacon.

The main parameters of the transceiver are reported in Table 10.2:

X-Band Tx operational frequencies	8.025-8.500 MHz
X-Band RX operational frequencies	7.145-7.250 MHz
Max transmitting data rate	25 Mbps
Optional Configuration Max data rate	100 Mbps
Max receiving data rate	64 Kbps
Power consumption	15 W (TX+RX) and 5 W (RX)
Mass	200 g
Dimensions	90 x 65 x 28 mm ³
RX modulation scheme	BPSK
TX modulation scheme	BPSK, QPSK, 8PSK
Ptx	2 W
Temperature range operation	-20°C, +50 °C

Table 10.2 - Main parameters of the X-Band transceiver [22]

The principal parameters of the antennas are reported in Table 10.3:

Operational frequencies	8.025-8.500 MHz; 7.145-7.250 MHz
Gain	11 dBi
Dimensions	60 x 40 x 1.8 mm ³
Mass	20 g
Temperature range operation	-30°C, +60 °C
TRL	9
Half power beamwidth	40°

Table 10.3 - Main parameters of the X-Band antennas [23]

The manufacturer has provided the chance to use an optional configuration of the X-Band transceiver with a maximum data rate of 100Mbps. It can also be added a Power Amplifier, in order to reach higher power transmissions.

For the preliminary analysis, a Power Amplifier available on the market was selected: it can reach up to 4W of RF Output power [24].

Then, the need of a beacon has been inserted as a requirement. In case of a failure of the mission or a failure of the TTMTTC subsystem, this device will be activated and will send a signal, in all directions, with the telemetry data of the CubeSat.

10.3 PRELIMINARY DESIGN AND PERFORMANCE EVALUATION

The link budget between Rhadamanthus and Europa Clipper (EC), considering the worst-case scenario, was evaluated as first step. In this phase, Rhadamanthus will transmit data and EC will receive.

It is known that Europa Clipper will perform multiple flybys of Europa, and during each of these fly-bys EC will orbit Jupiter in order to transmit the acquired data. Therefore, this last case will be the worst-case scenario, as it has the highest distance link.

The link distance in this scenario will be the distance between Jupiter and Europa:

$$d = 672560 \text{ km}$$

The transmitting frequency is: $f_{TX} = 8424 \text{ MHz}$

The distance loss was evaluated with the following formula:

$$L_s = \frac{4\pi fd}{c} = 227.5 \text{ dB}$$

A value for pointing loss was assumed: $L_s = 0.03 \text{ dB}$

The QPSK modulation was selected from the ones available from the transceiver.

This modulation allows the bitrate to be halved, so it is useful for the mission purpose. Then, a forward error correction (FEC) was selected: the Convolutional Coding with a rate of $\frac{1}{2}$ and $K = 7$.

The system will operate in two different modes: data and telemetry.

For the data mode, a BER of 10^{-5} is needed.

Considering the QPSK modulation, in order to have a BER of 10^{-5} it is required that the signal

arrives at the receiving end with: $\frac{Eb}{N0} = 9 \text{ dB}$: this value can be obtained from Figure 10.2

A margin of 3dB is added, and a bit rate of $R = 5 \text{ Kbps}$ was selected to achieve the minimum value of $\frac{Eb}{N0}$ also considering the margin.

The Energy per bit over Noise was evaluated with the following formula [25]:

$$\frac{Eb}{N0} = \frac{P_{tx} L_s L_s G_{tx} G_{rx}}{k T_s R}$$

The equivalent noise temperature of the transceiver was not provided by the manufacturer, a value of a similar transceiver available on the market was considered in the computation [26]:

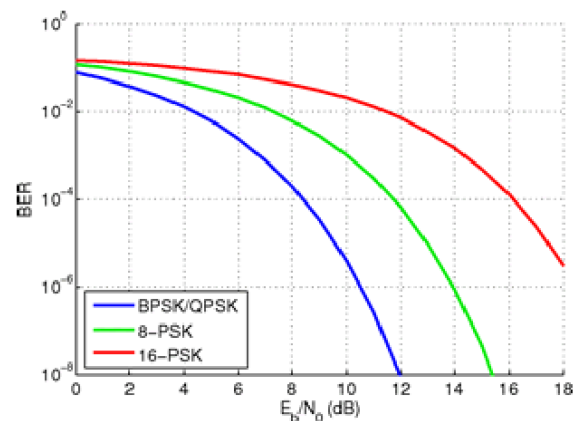


Figure 10.2 - BER vs E_b/N_0 [25]

$$T_S = 27 K$$

The same procedure was performed in to evaluate the performance and the needs of the system in telemetry mode, considering that this mode requires a BER of 10^{-6} and a $\frac{Eb}{N0} = 10.5$ dB.

To achieve the required quality of the signal, a bit rate of $R = 3.981$ Kbps is needed.

The link budget between EC and Rhadamanthus was then evaluated. In this phase, EC will transmit commands and Rhadamanthus will be receiving.

For this mode, known as commands mode, the operative frequency is: $f_{TX} = 7170$ MHz.

Distance losses were computed: $L_S = 226.2$ and a value for pointing losses were assumed: $L_s = 0.03$ dB.

The BPSK modulation was selected: it is the only one that the transceiver is capable to operate in receiving mode.

A value of $P_{TX} = 20$ W and a bit rate of $R = 15$ kbps were assumed since no information about the transmission power of Europa Clipper HGA was available.

A BER of 10^{-7} is needed for this mode, and then a signal with $\frac{Eb}{N0} = 11$ dB.

Mode	$\frac{Eb}{N0}$ [dB]
Data	12
Telemetry	13.5
Commands	14

Table 10.4 - Recap of Energy per bit over Noise values for each mode, considering a system margin of 3 dB

10.4 TIME WINDOW VISIBILITY

The strategy of the TTMTTC subsystem will be to operate with high data rate when Europa Clipper will perform flybys of Europa, and to transmit with the data rate evaluated in the previous chapter when EC will be orbiting Jupiter.

In order to have a first estimation of the time window visibility in the first case, the hyperbolic passage of EC was evaluated.

Europa Clipper will perform from 40 to 50 flybys of Europa for its science mission lifetime [27][28]: for this evaluation, a number of 45 flybys was selected. Europa Clipper flybys' altitude will vary from 2700 km to 25 km [27]: an average altitude of the hyperbola pericenter of $hp = 1362.5$ km was selected. It was assumed an eccentricity of the hyperbola of:

$$e_{hyp} = 1.84$$

The semimajor axis value is:

$$a_{hyp} = -3480.48 \text{ km}$$

Considering the half power beamwidth of the antenna $\theta = 40^\circ$, so $\theta^* = \pm 20^\circ$, and using the following formulas [29]:

$$\Delta t = \sqrt{\frac{-a^3}{\mu_{Europa}}} (t_f - t_i)$$

$$t = e \sinh(F^*) - F^*$$

$$F^* = 2 \tanh^{-1} \left[\sqrt{\frac{e-1}{e+1}} \tan \left(\frac{\theta^*}{2} \right) \right]$$

It was possible to evaluate the time window visibility:

$$\Delta t = 42.546 \text{ s}$$

Then, from simple computations, it was possible to consider that EC will perform approximately seven flybys of Europa during Rhadamanthus mission lifetime. So, for this case, the total transmitting time window is $T_{high R} = 297.8 \text{ s}$ and considering a bit rate of $R = 100 \text{ Mbps}$, the system is expected to send 29.7 Gbit.

For the worst-case scenario, so when EC will be orbiting Jupiter and the system will be operating with a bit rate of $R = 5 \text{ kbps}$, it was assumed that EC will orbit Jupiter with a circular orbit with an altitude of $h_{EC} = 75600 \text{ km}$ for simplicity: this value was the perijove altitude of Juno's mission orbit [30] around Jupiter. From this value is it possible to compute the period of EC orbit $T_{EC} = 8.6 \text{ h}$. Meanwhile, Rhadamanthus orbit is $T_{EC} = 2 \text{ h}$: for a rough estimation it is possible to assume that every four orbits of Rhadamanthus mission, it will be able to transmit for half the time of its orbit period to Europa Clipper: $T_{Tx} = 3757.87 \text{ s}$

Considering then the total number of Rhadamanthus orbits for which the mission will be able to transmit data, the total transmitting time window is: $T_{low R} = 2303611.88 \text{ s} = 26.7 \text{ d}$

For this case, the system is expected to send 11.51 Gbit of data.

11. THERMAL CONTROL SYSTEM

11.1 REQUIREMENTS

The thermal control system shall provide the thermal environment necessary to ensure full performance during each mission phase and mode as well as for the complete duration of the mission.

REQ. ID	IMPORTANCE	REQUIREMENTS
RHAD-TCS-010	M	The system's components shall be kept in the operative range.
RHAD-TCS-020	M	The system shall withstand thermal loads for the whole mission.
RHAD-TCS-030	M	The system shall survive radiation dose of the planets during the interplanetary transfer.
RHAD-TCS-040	M	The system shall be robust against aging phenomena and degradation of optical properties.

Table 11.1 – Thermal control system requirements

11.2 OPERATIVE CONDITIONS

The thermal control of the mission has the objective to maintain all the components within their allowed temperature ranges, reported in the Table 11.2.

Components	Temperature range (operative)	
	Min (°C)	Max (°C)
Electronics	-40	80
Propellant	2	113
Feedline / Tank	2	60
Antenna	-30	50
Transceiver	-20	50
Solar panels	-40	120
OBDH	-40	80
Scientific P/L	10	30
Batteries	-10	60
Reaction wheels	-20	70
ADCS sensors	-25	50

Table 11.2 – Operative temperature limits

The scientific payload (in particular the Chameleon Imager) imposes the more restricting conditions in terms of allowable temperatures. The minimum allowed temperature is 10°C (283 K), the maximum temperature allowed is 30°C (303 K).

11.3 ENVIROMENTAL INPUTS:

In the space environment, all the heat is transferred by radiation. The chosen coating is white painting, with absorptivity $\alpha = 0.2$ and emissivity $\varepsilon = 0.9$.

Possible heat sources:

1. Sun radiation
2. Albedo radiation
3. IR emission
4. Internal source

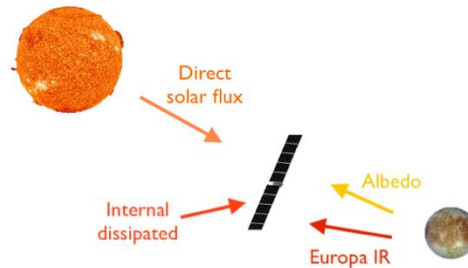


Figure 11.1 – Heat sources

1 - Sun Radiation

The solar flux depends on the distance of the planet from the Sun:

$$q_{sun} = 50 \frac{W}{m^2}$$

2 – Albedo Radiation

The albedo radiation comes from the reflection of the solar radiation on Europa's surface.

$$q_a = 25.5 \frac{W}{m^2}$$

Where the albedo of Europa is equal to 0.64.

3 – IR radiation

The heat coming from the IR radiation can be computed as:

$$q_{IR} = 13.5 \frac{W}{m^2}$$

Where the temperature of the planet is computed as follow: $T_{PL} = \sqrt[4]{\frac{(1-\alpha)q_{sun}}{\varepsilon\sigma}} \approx 125 \text{ K}$, assuming Europa as a slow rotating celestial body.

4 – Internal source

$$Q_i = 55.9 - 105.9 \text{ W}$$

The maximum and minimum values of internal dissipated heat are reported in Watts.

11.4 HOT AND COLD CASE SCENARIO

Final design: preliminary analysis assuming steady state conditions, modelling the spacecraft as a sphere with surface of $A = 0.66 \text{ m}^2$.

To perform thermal analysis, it's necessary to:

- Compute, from the environmental analysis, the values of heat from the 4 heat sources described above.
- Evaluate the worst hot and worst cold case.
- Select a proper coating material and select the number of layers needed to satisfy the main requirement of thermal control system.

Hot case: worst case, assuming the maximum value of the internal dissipated heat, albedo radiation, IR radiation and internal source.

$$Q_{hot} = Q_{i_{MAX}} + A(q_{IR} + q_a + q_s) = 165 \text{ W}$$

Cold case: worst case, assuming the minimum value of internal dissipated heat and IR radiation.

$$Q_{cold} = Q_{i_{MIN}} + A * q_{IR} = 65 \text{ W}$$

		Worst cold	Worst hot	Units
Sun heat flux	q_{sun}	0	33	W
Albedo radiation	q_{albe}	0	17	W
IR radiation	q_{IR}	9	9	W
Internal source	q_{int}	55	106	W
Total		65	165	W

Table 11.3 – Worst hot and worst cold cases

11.5 MULTI LAYERS INSULATION

The passive thermal control system is composed of a multi-layer insulation, with Aluminized Mylar. The spacecraft is assumed as a sphere, with surface area $A = 0.66 \text{ m}^2$.

In the MLI model, alternate layers of material are employed to protect the instruments and the spacecraft from extreme heat and are out of reciprocal contact to ensure radiation heat exchange only. $GR^* = A\varepsilon^*$, and $\varepsilon^* = \frac{1}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1} \left(\frac{1}{N+1} \right)$.

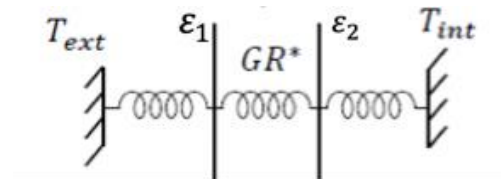


Figure 11.2 – MLI model

For the MLI, 30 layers of aluminized Mylar have been implemented to successfully guarantee the operative temperature range with a margin of $\pm 5 \text{ K}$. This coating layers will have a total thickness of about 1.5 cm. The effective emittance is equal to $\varepsilon^* = 0.04$.

The number of layers already contain a 25% margin after tests and a margin for the degradation due to the radiations. Considering the short time span of the mission and the material chosen, less than 7 months, the degradation in terms of solar absorptivity increase $\Delta\alpha$ is equal to $\Delta\alpha \sim 0.01$, which is sufficient to guarantee a thermal protection with the desired design maturity and system level margin.

12. ON BOARD DATA HANDLING

SPACECLOUD iX5-100 On Board Computer has been selected. It is under testing on D-Orbit Wild Ride ION Mission and will be flying aboard the NASA's Hyperspectral Thermal Imager (HYTI) mission.

The processing core is **e2160 Qseven compute module**, featuring a quad-core AMD CPU and AMD HD RADEON GPU.

This combination is expected to perform efficiently as long as it was designed specifically for hyperspectral imaging data handling.

The OBC interfaces for sensor readout and payload downlink with integrated S-band and X-band radios, allowing up to 100 Mbps data transfer.

In case the adopted data handling and downlink strategy allows to maintain fixed the maximum transfer rate until the whole storage data is uploaded to Europa Clipper, external SSDs interface is required. On the other hand, the integrated memory on Chameleon Imager and the expandable memory on the OBC provide up to 500 GB of internal storage together, which is enough if the selected downlink strategy limits the data volume by the short time available for upload during each fly-by of Europa Clipper.

Seagate IronWolf 125 Nas SSD 4TB is a SATA 3 device which has been chosen being one of the best products of the brand, most reliable with respect to its lifespan [32].

Mission Domain	IRONWOLF 125 4TB lifespan
Max data size: ≈ 115 TB	TBW: 5600 TB
Tof: ≈ 5119.2 h	MTBF: 1800000 h Max_ntof = 351

Table 12.1 – SSD lifespan parameters



Figure 12.1 – iX5-100 + e2160 Qseven [31]

Downlink when EC is visible
during each fly-by

$$Total\ Data\ Volume = \frac{t_{1DL} * R_{DL}}{t_{1rec}} * t_{totrec} = R_{rec} * t_{totrec}$$

Downlink when Rhadamanthus
approaches EC and visibility is constant

$$R_{rec} = \frac{Total\ Data\ Storage}{t_{rec}}; \quad t_{DL} = \frac{Total\ Data\ Storage}{R_{DL}}$$

where: $t_{(1)DL}$ = downlink time (during each Fly-by) $t_{(1)rec}$ = record time (before each Fly-by)
 R_{DL} = downlink bitrate R_{rec} = record bitrate

REQ. ID	IMPORTANCE	REQUIREMENTS
RHAD-OBDAH-010	NH	The system shall have a low power consumption mode
RHAD-OBDAH-020	M	External SSD shall be compatible through wired usb3 to sata3 adapter.
RHAD-OBDAH-030	M	Reliability test qualification of extSSD for space application (vibration, acceleration, deterioration of flash memory, reliability under radiation, etc.) shall be performed.
RHAD-OBDAH-040	M	Wired USB3 connection at docking point shall be operated mechanically (Rendez-vous case).

Table 12.2 – OBDAH requirement

13.PROPULSION SYSTEM

13.1 INTRODUCTION AND REQUIREMENTS

The Propulsion System (PS) on board of the CubeSat for Rhadamanthus mission is exploited to perform the insertion maneuver into the nominal orbit.

Once the CubeSat is released by Europa Clipper at the perigee of a hyperbolic arc at an altitude of 100km from Europa's surface, the thruster will be switched on to perform the desired maneuver.

In *Table 13.1* all the requirements regarding the propulsion system are listed.

REQ. ID	IMPORTANCE	REQUIREMENTS
RHAD-PS-010	M	The PS shall be able to guarantee the success of the insertion.
RHAD-PS-020	M	The PS shall bear a maneuvering cost of at least 0.685 km/s.
RHAD-PS-030	M	The provided thrust shall not make the CubeSat drifting in other directions except along the orbit.
RHAD-PS-040	M	The propellant mass shall satisfy the cost of the insertion maneuver.
RHAD-PS-050	M	The propellant shall not be subjected to undesired phase changes.
RHAD-PS-060	M	The PS shall not interfere with the Maspex.
RHAD-PS-070	M	The PS shall not interfere with the Chameleon Imager.
RHAD-PS-080	M	The thruster shall not be placed along the same axis of solar panels.
RHAD-PS-090	M	The propellant tank shall be placed in a position in order to be easily reachable from the thruster.
RHAD-PS-100	M	The pressurizing gas tank shall be placed in a position in order to be easily reachable from the propellant tank.

Table 13.1 - Propulsion system requirements

13.2 SYSTEM SELECTION

The first step of the sizing is the selection of the type of propulsion system among the currently existing ones: chemical, electrical, nuclear.

Nuclear propulsion has not been taken into account for the *Rhadamanthus Mission* for safety reasons and since it involves heavy systems, more related to long term missions.

Electric propulsion could be an adequate candidate, useful also in case of attitude correction maneuvers. Nevertheless, it implies more power consumption, which is relevant being in the Jovian environment; the Sun is not always accessible, it doesn't provide much power at around 5.2 AU and also the space allocated for batteries is limited.

Moreover, electric propulsion is not suitable for instantaneous maneuvers; for all these reasons a chemical system has been considered the most suitable for the mission.

Hybrid propellants provide good performance in terms of specific impulse and thrust, but they have a Technology Readiness Level (TRL) equal to 4, also there could be problems for what regards motor-scaling effects. On the other hand, the choice of solid propellants has been avoided, since the lack of information for what regards the combustion process would translate into a difficult and not accurate design of the fuel grain.

This reasoning leads to the choice of a liquid rocket motor, capable to provide good performance both in terms of specific impulse and thrust.

In particular a green monopropellant has been selected: AF-M315E [33].

The latter, known also as hydroxylammonium nitrate, is an ionic monopropellant which decompose into an oxidizer and a fuel, which then react together providing higher performance with respect common monopropellants.

In particular AF-M315E provide 10% higher specific impulse with respect to hydrazine, it is much less toxic than the latter, and also it is characterized by a high density (45% higher than N₂H₄) which lead to smaller and lighter propellant tanks, with a consequent lower wet mass. The thruster employed for the mission is the *Busek BGT-5* [34], whose main properties are listed in *Table 13.2*.

Nominal thrust	5 N	Thrust range	1 - 6 N
Propellant	AF-M315E	Propellant density	1470 kg/m ³
Specific Impulse	264 s	Power consumption	50 W
Heating chamber pressure	400 psi = 2.758 MPa	Mass	1.5 kg

Table 13.2 - Thruster specifications

As it can be seen from the previous table, the thruster spends a lot of power and the fact is due to the need of the system to pre-heat the catalyst bed, before the firing.

The thruster will be employed just at the beginning of the mission, to perform the insertion maneuver, for this reason the power consumption is admissible since the primary batteries at that stage are fully charged.

13.3 PROPULSION SYSTEM DESIGN

In this paragraph the main results of the propulsion system sizing will be presented.

For design purposes, a pressure-fed system based on CO₂ has been chosen; the architecture is shown in *Figure 13.1*.

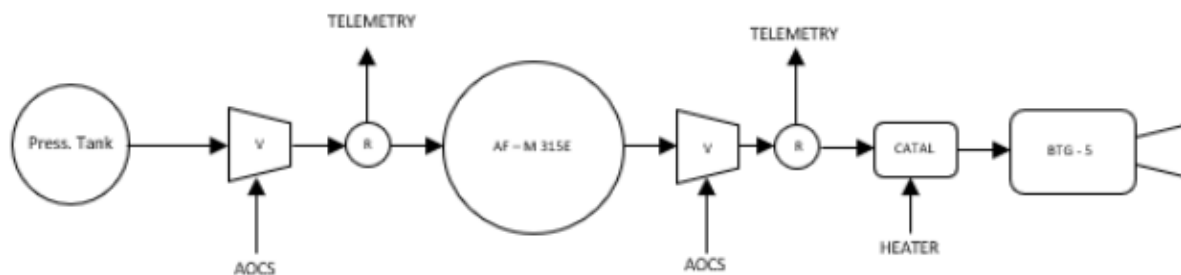


Figure 13.1 - Propulsion system architecture

13.3.1 PROPELLANT BUDGET

Once the needed ΔV to successfully accomplish the mission and some specifications given by the chosen propellant are known, it is possible to compute the necessary amount of propellant starting from the value of total dry mass at launch.

In particular, exploiting Tsiolkovsky's equation:

$$m_p = m_{dry} \left[\exp \left(\frac{\Delta V}{g_0 I_{sp}} \right) - 1 \right] = 11.635 \text{ kg}$$

13.3.2 TANKS SIZING

The propellant tank has been sized starting from the volume to store the propellant mass needed, adding a margin of 10%.

Considering also the space constraints given by the configuration of the CubeSat and the allocation of the other subsystems, a cylindrical tank with radius $R_{tank} = 0.11m$ and height $h_{tank} = 0.22m$ has been selected.

In order to determine the operating pressure of the tank, the following contributes have been considered:

$$P_{tank} = P_{cc} + \Delta P_{inj} + \Delta P_{feed} + \Delta P_{dyn}$$

The operating pressure of the heating chamber is given by the thruster specifications, the pressure losses given by the injection has been considered the 10% of the heating chamber pressure, while the pressure losses given from the feed lines are estimated as 0.05MPa, a typical value. Finally, the dynamic pressure losses have been evaluated as $\Delta P_{dyn} = 1/2\rho_p v_p^2$, where the propellant velocity has been considered equal to 10m/s.

This leads to an operating pressure of $P_{tank} = 3.157 MPa$ and a burst pressure of $P_{burst} = 6.314 MPa$.

Regarding the pressurizing gas, CO2 has been chosen for its lower specific heat ratio and its higher molar mass with respect to helium or nitrogen, to permit the storage in a smaller tank. The pressurizing gas tank features have been collected in *Table 13.3*.

Geometry	Cylindrical	Initial Pressure	45.284 MPa
Radius	0.054 m	Final pressure	3.157 MPa
Height	0.120 m	Mass of CO2	0.693 kg

Table 13.3 - Pressurizing gas tank properties

In order to establish the thickness and the mass of both tanks, three materials have been considered: aluminum alloy, carbon fiber and steel. The results are shown in *Table 13.4*.

Material	Propellant tank thickness	Pressurizing gas tank thickness	Propellant tank mass	Pressurizing gas tank mass
Aluminum alloy	2.2 mm	15.9 mm	1.385 kg	2.559 kg
Carbon fiber	0.77 mm	0.55 mm	0.274 kg	0.507 kg
Steel	0.81 mm	0.57 mm	1.439 kg	2.659 kg

Table 13.4 - Material choice for the tanks

The chosen material for both tanks is carbon fiber.

13.4 ALTERNATIVE ARCHITECTURE

For liquid monopropellant rocket motors, pressure-fed systems are not quite common. Moreover, this architecture, in this application, leads to store pressurizing gas in a small space, around 1U, at very high pressure, around 45 MPa, increasing the risk of explosion.

A suitable alternative could be represented by multiple blow-down systems, connected to the same feed line in order to correctly feed the thruster. This architecture implies a variable thrust in time and this fact shall be evaluated to do not compromise the insertion maneuver. This possible solution needs also the new evaluation of the CubeSat's configuration, to determine the correct number of components and their allocation into the spacecraft.

14. ELECTRIC POWER SUBSYSTEM

14.1 INTRODUCTION

The Electrical Power Subsystem is in charge to convert any energy source into electrical energy; the architecture shall be able to provide a continuous electrical power and support power peaks. It is needed to develop an energy storage to use it whenever the primary source is unavailable.

The electrical power must be distributed to each part of the spacecraft where the electrical energy is needed, nevertheless an intelligent command and control system is needed to perform an efficient job.

14.2 REQUIREMENTS

REQ. ID	IMPORTANCE	REQUIREMENTS
RHAD-EPS-110	M	The CubeSat shall obtain enough power able to provide enough electrical energy to run all the instruments and payloads during the life mission.
RHAD-EPS-100	M	The EPS shall charger and priority power setting of payload and battery.
RHAD-EPS-110	M	The EPS Shall recharge the secondary batteries once the primary source is available.
RHAD-EPS-130	M	The EPS shall change from the generator to the secondary batteries during eclipse.
RHAD-EPS-140	M	The Solar panels shall be folded until the CubeSat stay in position to start orbiting.

Table 14.1 – Electrical Power Subsystem Requirements.

14.3 POWER BREAKDOWN

In order to design a proper EPS, it has been summarized the total amount of power required for each component known.

Area	Instruments	Budget Breakdown	Cruise	Eclipse	Sunlight
Control	Computer	30W	10W	30W	30W
Propulsion	Thruster	50W	50W	0W	0W
TT&C	Antenna Tx	14W	14W	14W	14W
	Antenna Rx	5W	5W	5 W	5W
ADCS	ADCS Board	3W	0W	3W	3W
	Reaction wheel	4.5W	0W	4.5W	4.5W
Payloads	Imaging System	7W	0W	7W	7W
	MASPEX	6.4W	0W	6.4W	6.4W
Total Power		119.9W	79W	69.9W	69.9W
Power +30% of Margin		155.87W	102.7W	90.87W	90.87W

Table 14.2 – Power Breakdown.

As it's shown in the table above the total Electrical Power Budget Breakdown its segmented in 3 parts:

- Cruise: this only refers to put in orbit Rhadamanthus Cube Sat once has been detached from Europa Clipper that means the operating instruments consuming energy are the one which are able to support this maneuver and the ones that provides communication with the ground station.
- Sunlight: is a segmentation when the primary source of light is available.
- Eclipse: is when the sun lights its blocked for another body; in this case the primary source of energy based on a photovoltaic reaction is not more available.

It's important to consider a power system level margin in order to assure the mission success and support any peak of electrical energy demand.

14.4 POWER SOURCE

The power source has been divided in two sources: photovoltaic solar panel and primary batteries.

14.4.1 SOLAR PANELS

For the purpose to know the solar panels area it's needed to compute the total amount of power supplied for the solar panels during the sun light and eclipse in the worst-case scenario. According with the mechanical orbits, we can compute what would be the time in sun light and in eclipse applying the next formula and the information in *Figure 14.4.1* [35].

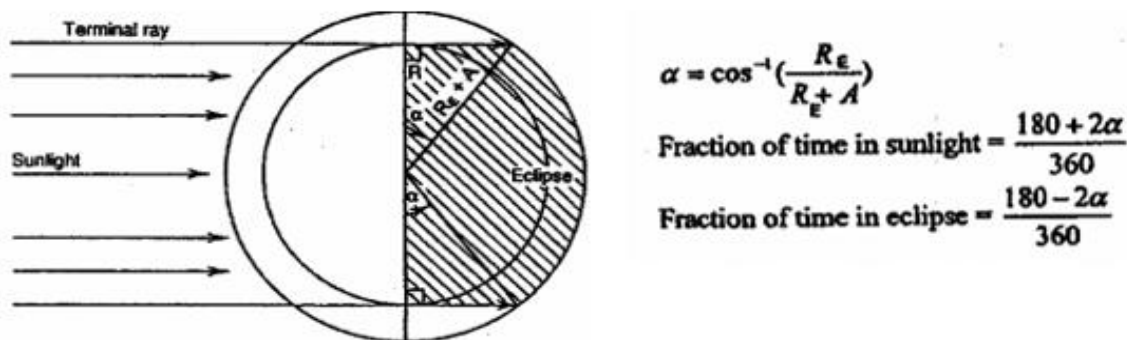


Figure 14.1 – Fraction of time

	Europa - Jupiter	Rhadamanthus - Europa
Jupiter Radius	69 911 km	-
Europa Radius	-	1 560.8 km
Orbital Radius	670 900 km	1 660.8 km
Orbital period	85.224 h	2.08 h
α	84°	20°
Sunlight	82.337 h	1.27 h
Eclipse	2.838 h	0.81 h

Table 14.3 – Orbit's data

The worst-case scenario will be after 40 orbits when Rhadamanthus face the two eclipses and it has been supposed one comes subsequently the other, due to this phenomenon we sum both eclipses.

$$\text{Europa's eclipse} + \text{Jupiter's eclipse} = 0.81 \text{ h} + 2.8 \text{ h} = 3.61 \text{ h}$$

Then the total power of the solar panels is computed using a DET power regulation control (Direct Energy Transfer) the reason is defined since it is a more efficient design, simple, light, used with a numerous eclipse with a rapidly change, then the next constants are defined as $X_e=0.65$, $X_d=0.85$. Nevertheless, this case must be studied in detail doing the electrical circuit.

$$P_{SA} = \frac{\frac{P_e T_e}{X_e} + \frac{P_d T_d}{X_d}}{T_d} = 504.3 \text{ W}$$

$$\text{Where: } P_e = P_d = 90.87 \text{ W}, T_e = 1.27 \text{ h}, T_d = 3.61 \text{ h}, X_e = 0.65, X_d = 0.85.$$

After that it is needed to know what is the total amount of energy that can be obtained in this system.

$$P_{BOL} = P_0 I_d \cos(\alpha) = 41.34 \frac{\text{W}}{\text{m}^2}$$

P_0 is the specific power generated in W/m^2 at 1 AU= 50 W/m^2 .

I_d = is an ideal assumption of the inherent degradation = 0.88

α it is an ideal assumption of the sun light impact angle = 20°

Due to the low value of the power achieved by the solar panels our EPS demand the use of a concentrators in order to achieve a higher efficiency. An SLA (Stretched Lens Array) concentrator has been selected and the efficiency increase up to 29% [36].

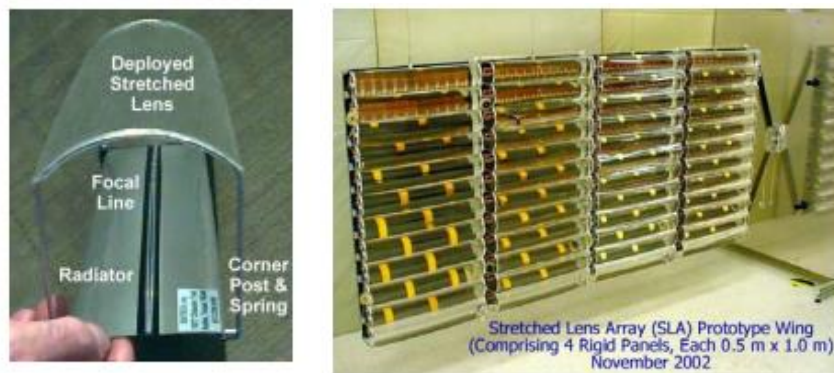


Figure 14.2 – Stretched Lens Array

$$P_{BOL} * \eta_{SLA} = P_{BOL} * 1.29\% = 53.33 \frac{\text{W}}{\text{m}^2}$$

Then we add a degradation factor L_d due to the losses during time this factor is obtained according with theory where after 6 years there a degradation of 15%; In this mission it's

expected to fly for 7 months, for this reason is taken a degradation of 1 year equal to 2.5%. After that the power produced at the end of life can be computed.

$$P_{EOL} = P_{BOL} * L_d = 53.33 \frac{W}{m^2} * 0.975 = 52 W/m^2$$

Finally, the total area of the solar panels is computed as the P_{SA} in the worst-case scenario divided by the power provided by the solar panels per m^2 .

$$A_{SA} = \frac{P_{SA}}{P_{EOL}} = 9.69 m^2$$

14.4.2 PRIMARRY BATTERIES.

It is needed a second power source since there is a power demand on cruise phase when the solar panels are not operating and there is a demand of 102.7 W, as is described in *Table 14.3*.

For this reason, had been selected the use of primary batteries, specifically Lithium Polymer Batteries which offer high capacity with low mass and volume. The one used are OPTIMUS 80 and OPTIMUS 30 connected in series to provide a total amount of power needed.

These special batteries are used for LEO missions that had been tested by NASA and ESA. Another important point is the battery systems have an autonomous integrated heater to enhance operation at low temperatures [37].



Figure 14.3 – OPTIMUS Batteries

14.5 STORAGE

The spacecraft shall store electrical energy and the solar panels are able to generate more energy in order to satisfy the requirements when the primary energy is unavailable.

The total amount of charge and discharge are strongly related with the total number of cycles equal to 2452 orbits.

- Total time on sunlight= (Number of cycles * time in sunlight) - (60 Jupiter eclipses with 20.028 h) = (2452*1.27)-121.7 = 2992.36 hours.

- Tot time on eclipse = (Number of cycles * time in eclipse) + (60 Jupiter eclipses) = (2452*0.81) +121.68 = 2107.8 hours.

Note: Each eclipse of Jupiter takes just 2.028 h of sunlight

After 41 Rhadamanthus orbits through Europa face 1 eclipse of Jupiter therefore the mission face 60 Jupiter eclipses.

Due to the high value of Cycles with a Lithium Polymer Battery we have a DoD = 58%

Then the Energy power provided by the secondary batteries can be computes as:

$$E = P_e T_e = 90.87 \text{ W} * 3.61 \text{ h} = 328 \text{ Wh}$$

At the end it has been selected 5 Pack 8 cells LifePo4 batteries with a capacity of 84 W/h per battery giving a total of 420 W/h [38].

14.6 POWER CONTROL

The EnduroSat EPS Type II has been selected as mainly controller; it has an aluminium housing that improves the thermal capabilities of the whole system, reduces the EMC, and protects the electronics from radiation.

It shall control the state of the self-lock function, power BUS, charger and priority power setting of payload and batteries.

14.7 POWER DISTRIBUTION

In this high-level approach, it is not designed the electrical circuit but is taking in account the impact of the main bus, electrical harness and terminal into the mass budget, assuming a 15% of the EPS equal to 0.5 Kg

15. MASS BUDGET

15.1 INTRODUCTION AND REQUIREMENTS

The mass budget is necessary to compute the thrust needed for the orbital maneuvers performed.

REQ.ID	IMPORTANCE	REQUIREMENT
RAD-STR-D-190	M	The CubeSat must have a rigid structure where all the instruments be fitted.
RAD-TMTC-D-020	M	The CubeSat shall have a telecommunication system.
RAD-PS-D-040	M	The CubeSat shall have a monopropellant propulsion system
RAD-EPS-D-060	M	The CubeSat shall have a primary unit of power distribution.
RAD-EPS-D-070	M	The CubeSat shall have a secondary unit of power distribution.
RAD-ADCS-D-080	M	The CubeSat shall have ADCS on board.
RAD-TMTC-D-120	M	The CubeSat shall have two antennas, one for receive data and the other one to transmit data.

Table 15.1 – Mass Budget Requirements.

15.2 MASS BUDGET BREAKDOWN

In Table 15.2 are listed all the components with the respective mass. 5% of margin for off-the-shelf components without modification and 10 % if they are modified.

Area	Instrument	Mass Kg	Safety Margin	Final Mass Kg
STRUCTURE	24U CubeSat structure	2.500	5.00%	2.625
PAYLOAD	Imaging system	1.600	10.00%	1.760
	MASPEX	10.000	10.00%	11.000
EPS	EPS system + Battery Pack	4.405	5.00%	4.625
	Solar Panels	9.000	10.00%	9.900
OBDH	Computer	0.500	5.00%	0.525
	ONC + SSD	0.310	5.00%	0.326
TMTC	Antennas	0.04	5.00%	0.042
	Transmitter + receiver	0.256	5.00%	0.2688
PROPULSION	Propulsion system	2.281	10.00%	2.509
	Propellant mass	11.635	10%+2%	13.054
ACDS	ADCS board	0.450	5.00%	0.473
	4 Reaction wheels	4.000	5.00%	4.200
TCS	MLI	0.150	10.00%	0.165
	TOTAL DRY MASS	35.492	-	38.418
	TOTAL MASS	47.127	-	51.472

Table 15.2 – Mass Budget Breakdown.

It is important to underline that most of the components should be manufactured; for this reason, it should be safer to consider a margin of 10% on each instrument. Moreover a 20% system level margin on the total dry mass value has to be considered.

The consequences will change the total mass of the propellant needed as well as the configuration in order to fit all the subsystems in our requested volume.

16. STRUCTURE AND CONFIGURATION

16.1 INTRODUCTION

The spacecraft design and configuration shall follow the functionalities of each payload and instrument in order to accomplish the mission; it is needed to analyze different configurations from different point of views and choose the best feasible configuration.

16.2 REQUIREMENTS

REQ.ID	IMPORTANCE	REQUIREMENT
RHAD-STR-010	M	The total volume of the CubeSat shall fulfill the constraint of 24U.
RHAD-TTCM-030	M	The CubeSat' antenna shall be positioned on the opposite side from the thruster to avoid perturbances.
RHAD-EPS-090	M	The CubeSat shall have deployable solar panels.
RHAD-ADCS-D-160	M	The set of 4 reaction wheels shall be positioned in a tetrahedral configuration.
RHAD-PS-D-170	M	The propellant tank shall be cylindrical.
RHAD-PS-D-180	M	The pressurized gas tank shall be cylindrical.
RAD-STR-C-020	M	The imager system shall point Europa
RAD-STR-C-030	M	The MASPEX shall point at the orbit direction
RAD-PS-C-030	M	The thruster shall be placed in the bottom of the CubeSat.
RAD-EPS-C-050	M	The EPS system shall be placed in the center of the CubeSat to easily feed all subsystems.
RAD-TMTC-C-060	M	The two antennas shall be placed outside the spacecraft, where they are not perturbed by solar panels.
RAD-PS-C-070	M	The pressurized gas tank shall be placed in a position easily reachable from the propellant tank.
RAD-STR-C-080	M	Solar panels, once deployed, shall not interfere with the imager field of view.

Table 16.1 – Structure and configuration.

16.3 VOLUME BREAKDOWN

In the *Figure 16.1* it is described the volume of the payloads and instruments and an exploded assembly.

There are two component that goes out of the 24 U constraint:

- The antennas, both located on the top side of the CubeSat. The maximum thickness of one of the antennas is 2.8 mm as it's described in Chapter 10, and both are assembled out of the spacecraft; for this reason, the antennas are not considered into the volume constraint.
- The solar panels have a large area nevertheless they're not considered in the volume constraint according with the customer requirements.

Component	Unit
Chameleon Imager	2U
MASPEX + EBOX	8U
Computer	0.6 U
EPS system + Battery Packs	2 U
ADCS board	1U
4 Reaction wheels	1U
Antennas	0.02 U
TTMC Control	0.4 U
Propulsion system	9 U
<i>Solar Panels area</i>	<i>9.6 m²</i>

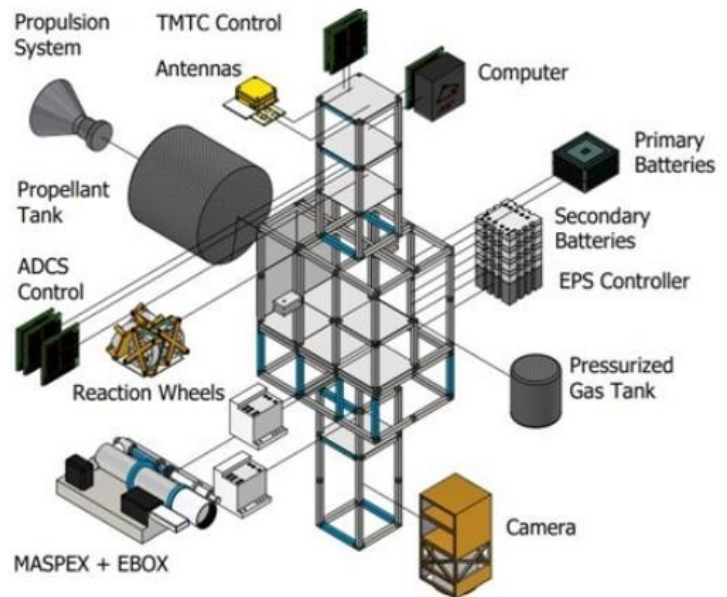


Figure 16.1 – Volume Breakdown

16.4 CONFIGURATION

As is requested some functionalities depends on the position of the instruments therefore this becomes the main driver to design the Cube Sat. Here you can find the main relations:

- The imager is looking Europa' surface in Nadir vector and the antennas are in the same axis looking the depth space.
- The propulsion system is aligned to the velocity center of mass to provide a uniform thrust.
- The ACDS subsystem is close to the CoG to perform an easy slew maneuver
- The MASPEX face the direction of the velocity vector against the emission of the propulsion system
- The Solar Panels are looking for the sun and the batteries surface are in the opposite surface.

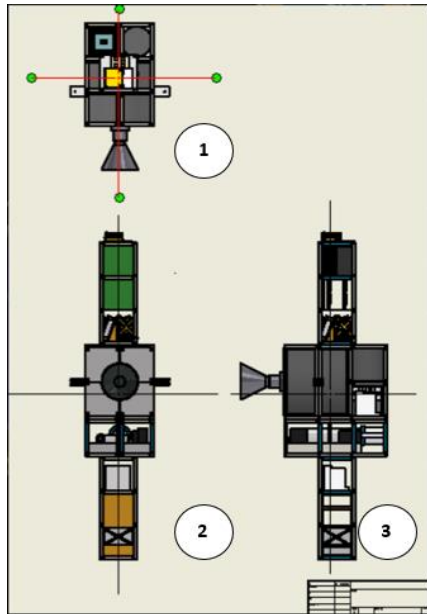


Figure 16.2 – 1 Top view, 2 Back view, 3 Right side view

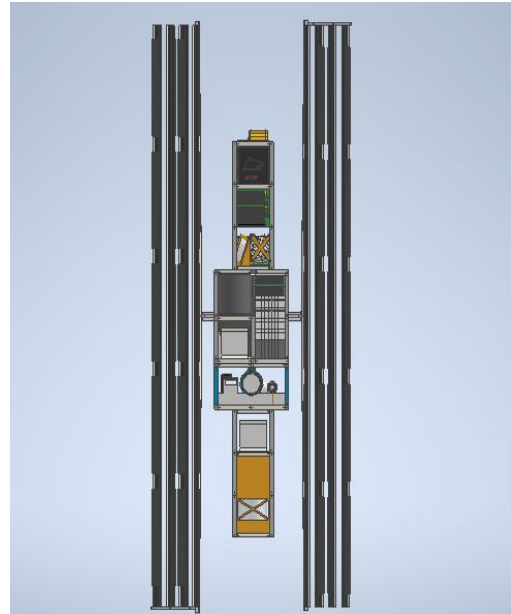


Figure 16.3 – CubeSat with Solar panels folded

16.5 OPPORTUNITY AREAS

This design and configurations work according with the requirements, nevertheless there are critical points to modify and improve:

- This configuration should be analyzed from the mechanical point of view since, due to the geometry, it could exist strong load stresses and vibrations which affects the main structure and the junction points.
- The dimensions of the solar panels are critical since they are 32 times longer than the CubeSat (*Figure 16.4*).
- Blowdown architecture with multiple tanks should be analyzed to get an efficient use of space



Figure 16.4 – Front view of Rhadamanthus CubeSat

17.CONCLUSIONS

17.1 INSTRUMENTS AND PAYLOAD TECHNOLOGY READINESS LEVEL TRL

The Technology Readiness Level (TRL) is an index that allows to measure the state of maturity of the technologies. It identifies the phases that start from the conceptualization of a new technology up to its effective use in a space mission [39].

For the two scientific payloads employed in the Rhadamanthus mission, the TRL are the following:

Instrument	TRL
MASPEX	6
Chameleon imager	7

Table 17.1 – TRL of payloads

The MASPEX is suitable for applications to Discovery, New Frontiers and Flagship missions [5]. The Chameleon imager functionalities are demonstrated in space environment.

17.2 OPEN POINTS

Different open points must be addressed to successfully accomplish the mission.

The main problems that have to be solved regard:

- End of life phase: the possibility of a docking with Europa Clipper at the end of the mission shall be evaluated. In any case, the requirements about the planetary protection for the category III/IV (like the use of clean rooms during assembly and inventory of bulk constituent) must be verified.
- Mission risk: According with the mothership, Europa Clipper shall be launched in October 2024; this requires to optimally manage available resources to obtain materials purchase, laboratory research, specialized personnel to reach Rhadamanthus mission qualification since there are still open points to clarify.



3 Strategies for uplink to Europa Clipper				
	Data Volume	Compressed Video Data Rate	Time for transfer	#
Send data to EC in low visibility	$\leq m \cdot 1.44 \text{ GByte}$	$\leq m \cdot 0.625 \text{ kbps}$	$m \cdot \text{tof}$	$m = \text{number of repetitions of full original mission (tof)}$
Send data to EC when it is in good visibility	$\leq m \cdot 3.41 \text{ GByte}$	$\leq m \cdot 1.48 \text{ kbps}$		
Approach EC and orbit in constant ($\leq \text{opt}$) visibility	$n \cdot \text{SC (4TB)}$	$n \cdot 1.73 \text{ Mbps}$	$\geq n \cdot 3 \text{d}17 \text{h}$	$n = \text{number of SSDs}$ $\text{SC} = \text{SSD Storage Capacity}$
RDV with EC	$n \cdot 4 \text{TByte}$	$n \cdot 1.73 \text{ Mbps}$	$n \cdot 2 \text{h}13 \text{min}$	

Table 17.2 – Uplink strategies

17.3 – CRITICALITIES AND POSSIBLE SOLUTIONS

In the following table, the different criticalities detected during the preliminary feasibility analysis are reported:

Area	Criticalities	Possible solution
Mission Analysis	Orbit insertion and final maneuver to exit from Europa SOI	Increase the size of the spacecraft to increase the mass of propellant and the Δv budget
	End of life phase	- Cleanrooms shall be used during spacecraft assembly and testing - An inventory of bulk constituent organics is required - The possibility of a docking with Europa Clipper at the end of the mission shall be verified
ADCS	Maximum slew rate	Maximum slew rate shall be verified in order to do not exceed 50% momentum storage in reaction wheels.
	RW de-saturation	Add cold gas thrusters, the CubeSat's configuration shall be revisited.
TTMC	Short Time window visibility	Need to develop high gain antennas and reach high power transmissions;
	- High science data amount to transmit - High link distance - High total time required to transmit all acquired data	- At the end of the mission, if enough propellant is available, orbiting Jupiter will permit to increase the data rate and the time visibility with EC, increasing total transmitted amount of data; - Re-design of telecom transmitting strategy, leaving some parts of Europa's surface not mapped.
TCS	Non-uniform distribution of internal dissipated heat	Implement active thermal control devices in specific places, e.g. radiators

Area	Criticalities	Possible solutions
OBDH	• High power consumption • Qualification of extSSD • RDV case: link to EC at docking	• Research for low power alternatives or modes. • Electric motor which operates USB connection at docking point.
Propulsion	• Pressurizing gas tank	• Adopt multiple blow-down systems.
EPS	• Low sensitivity to sun	• Increment area of solar panels. • Motion of solar array to optimize the angle of impact. • Concentrator SLA
Structure Configuration	• 24 U constraint • Payloads and equipment pointing. • Propulsion system and ADCS alignment. • Tanks Design. • Spacecraft geometry	• Create multiple versions constrained by the functionalities. • Reaction wheels shall stay close to the CoG. • Geometric variables. • Modify the s/c geometry keeping a symmetrical shape.

Table 17.3 – Mission criticalities and possible solutions

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